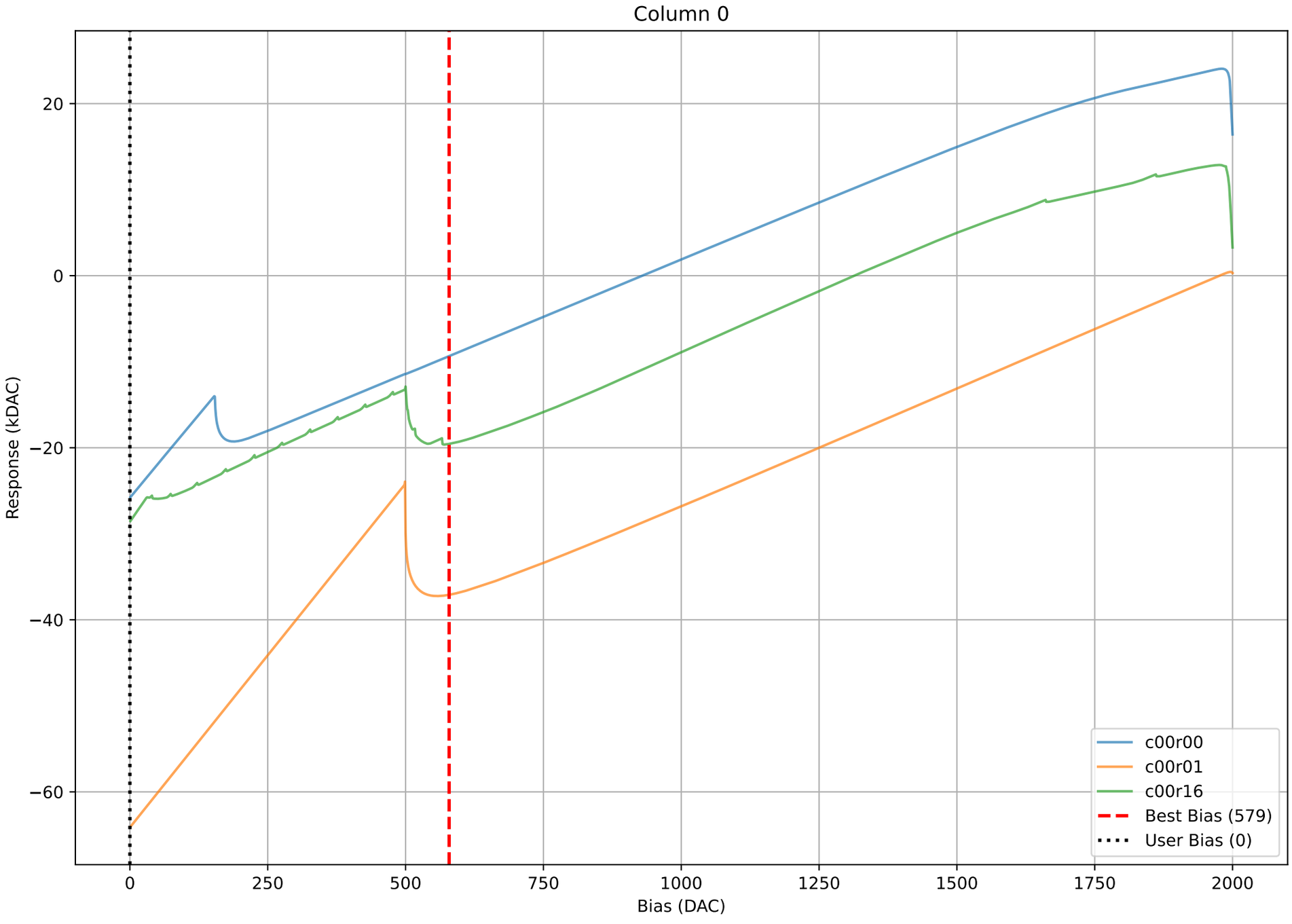
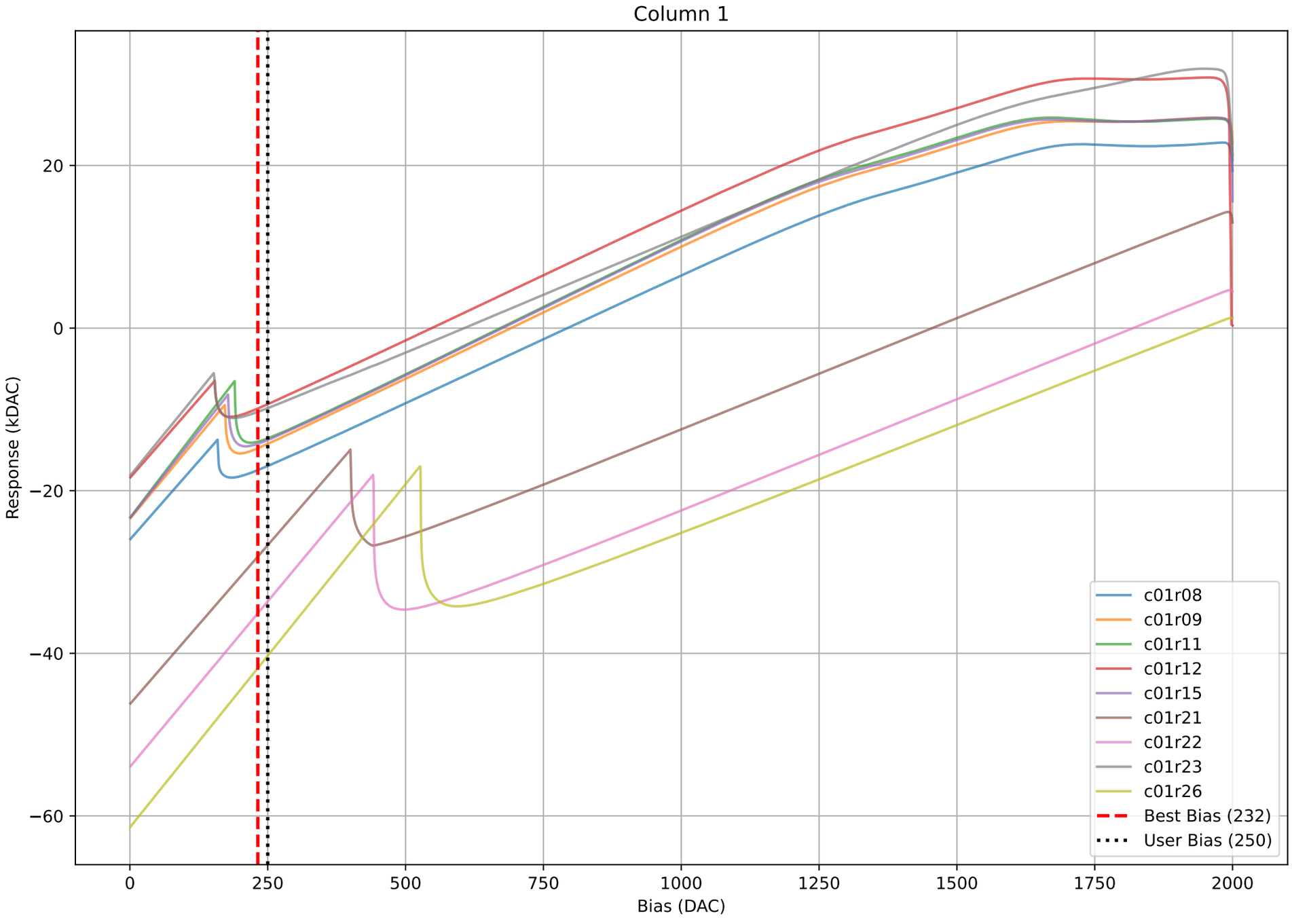
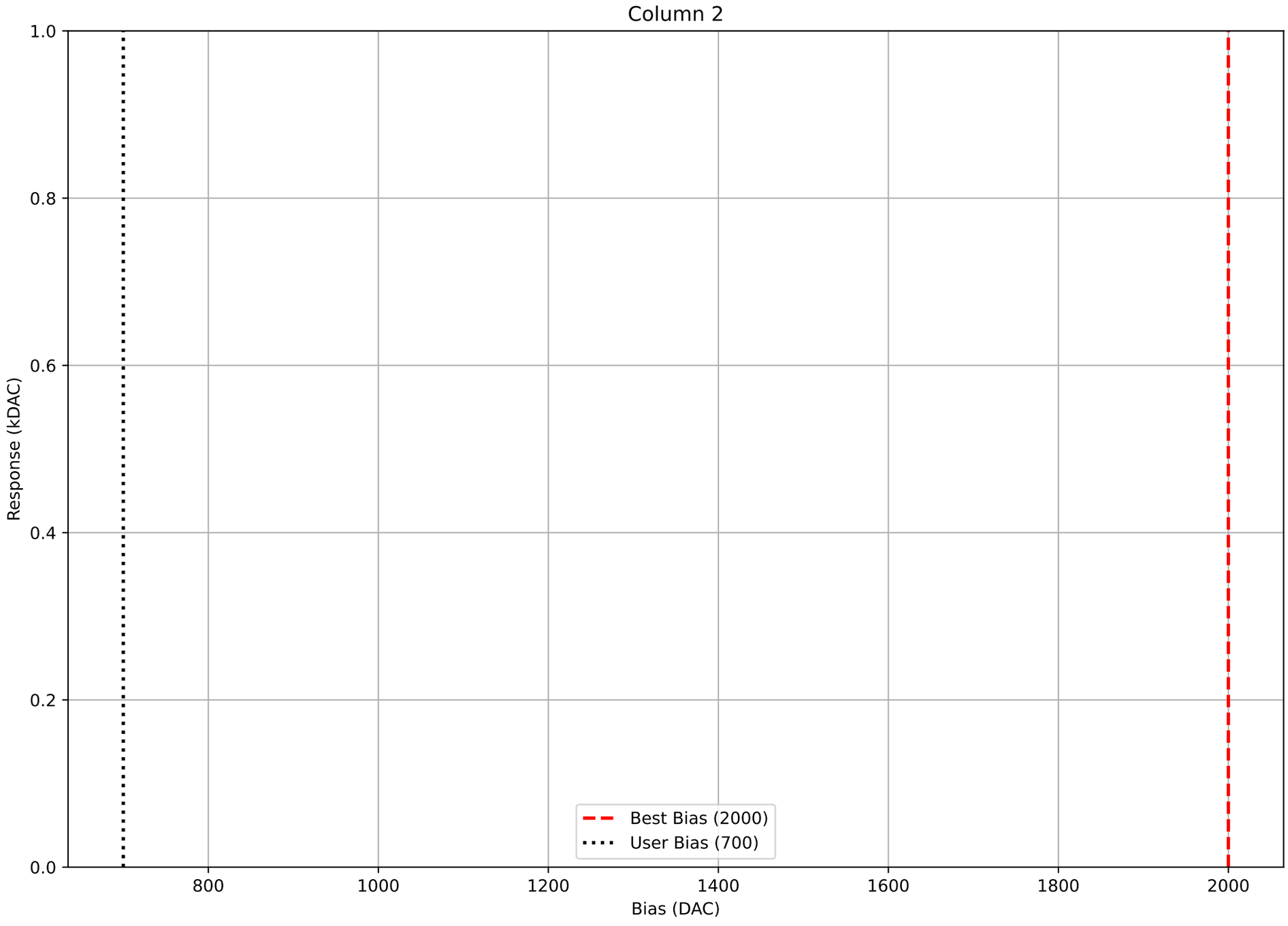
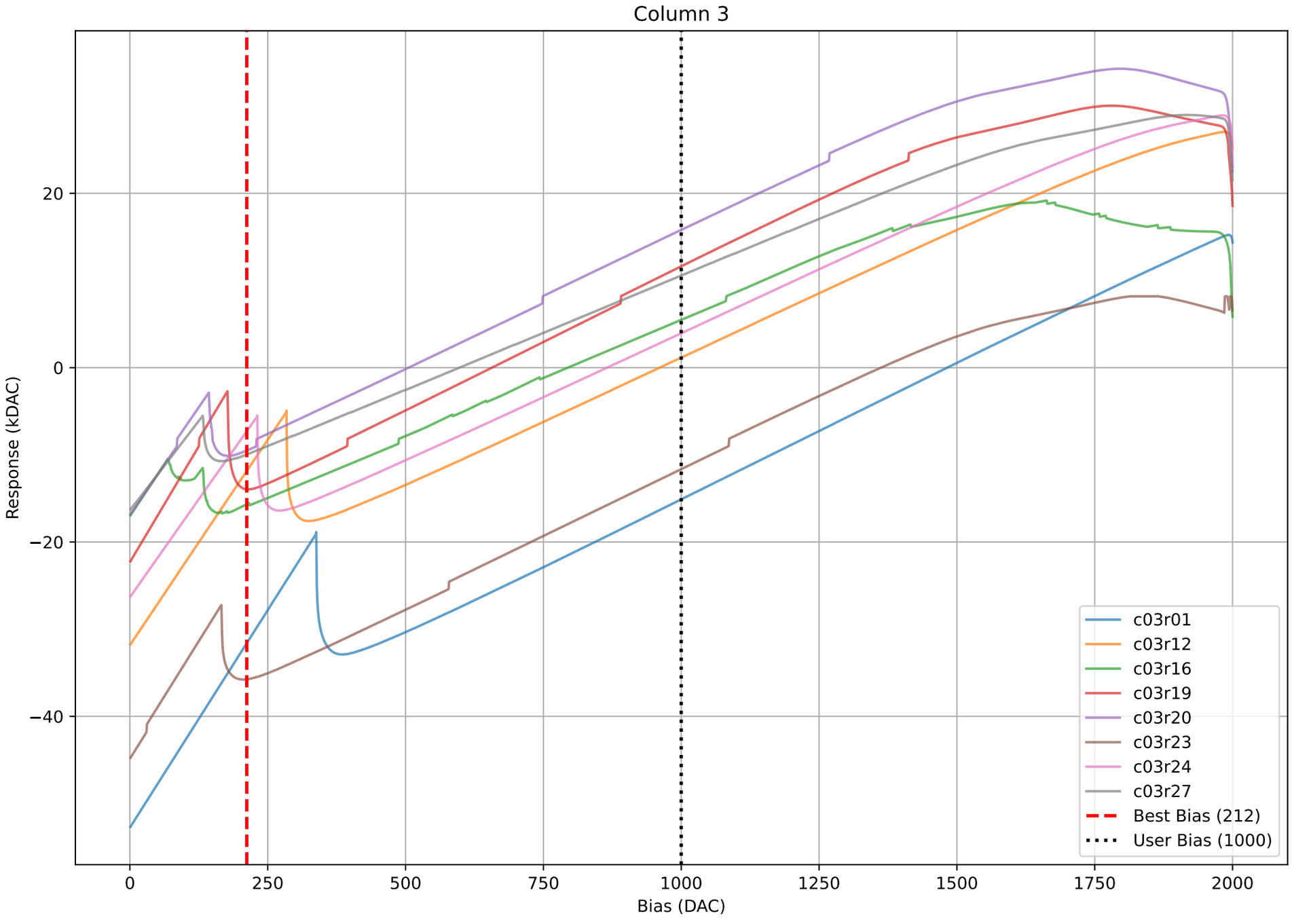
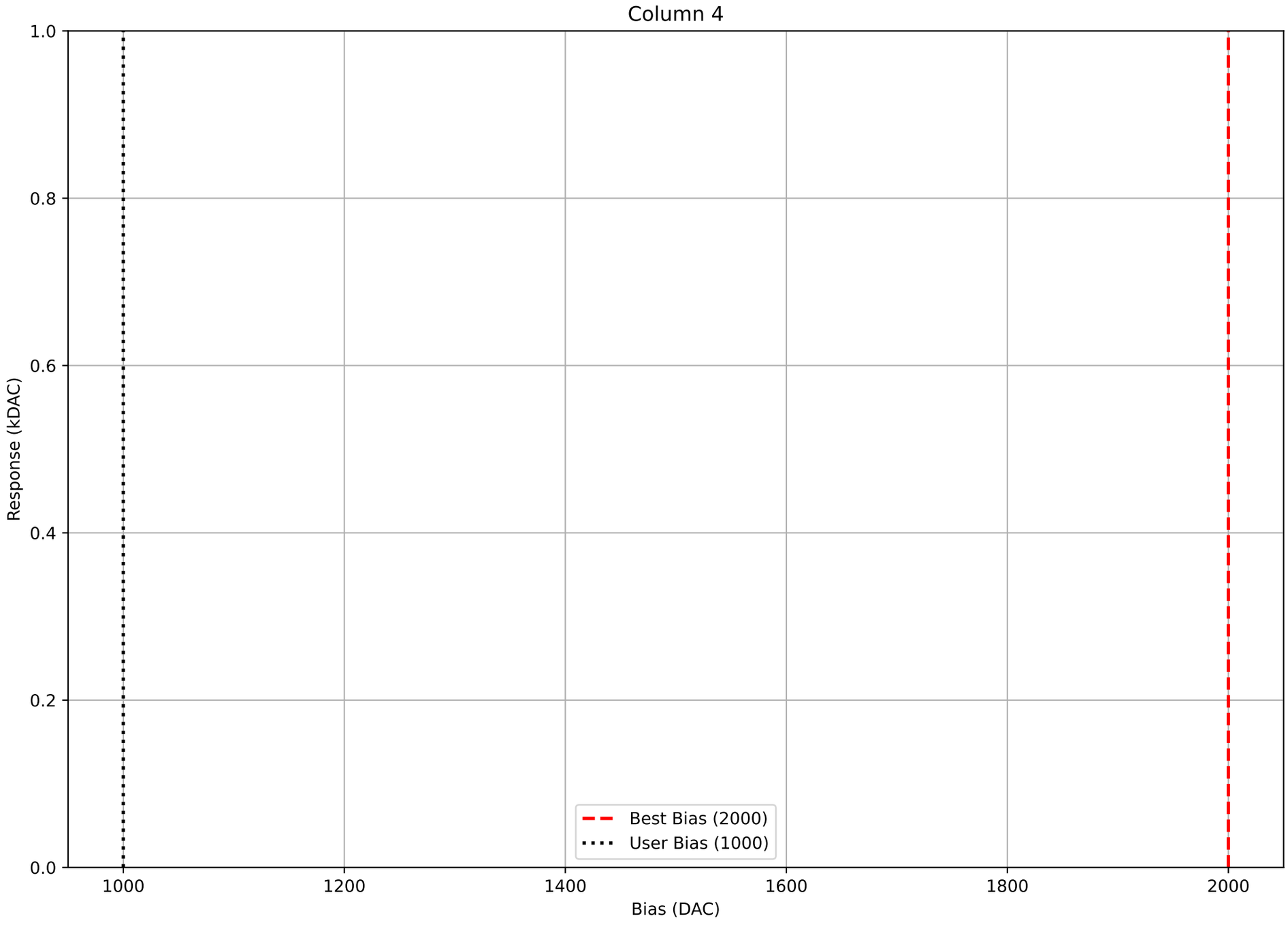
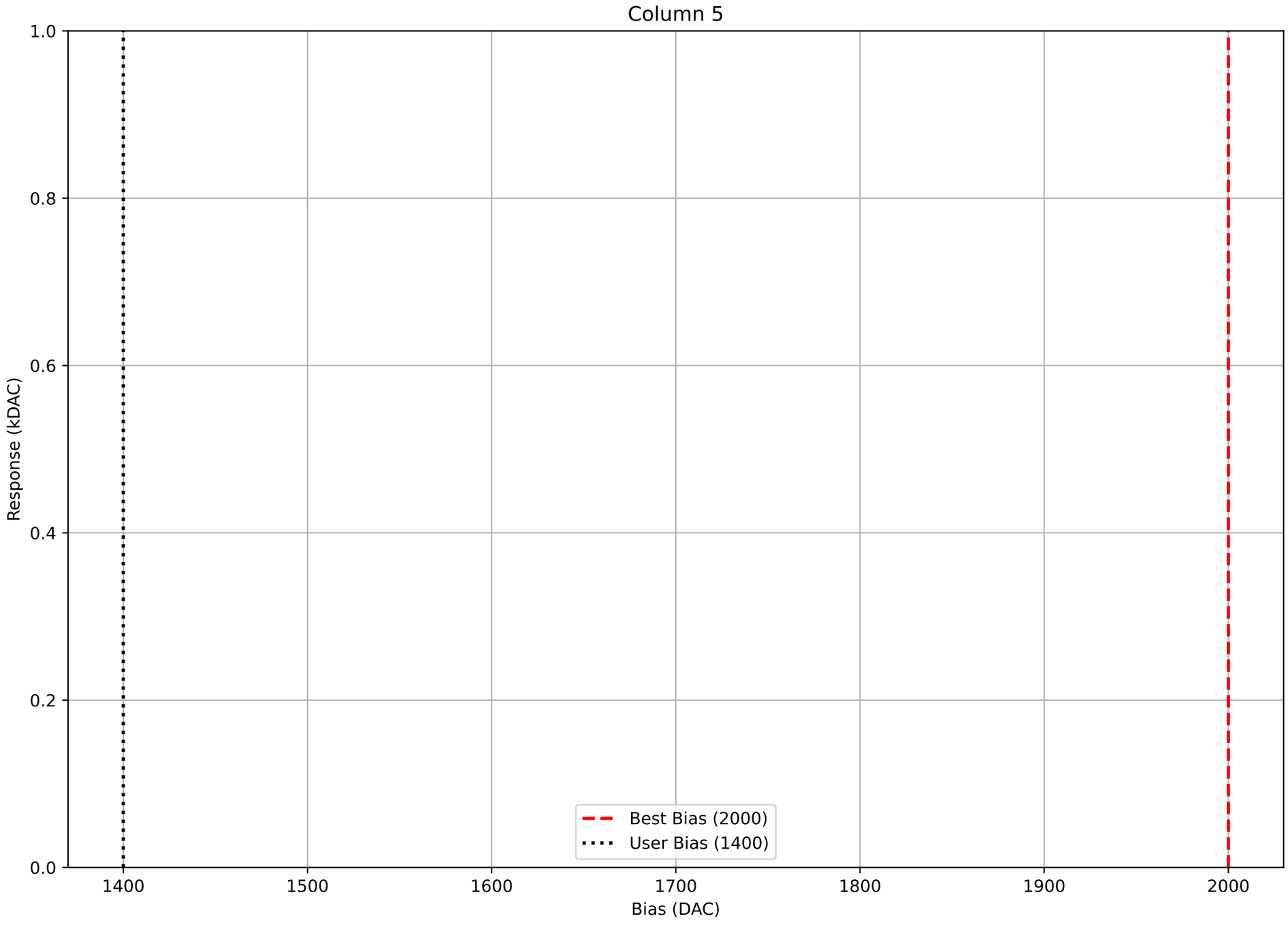

Column 0
Response (kDAC)
20
0
−20
−40
−60
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
c00r00
c00r01
c00r16
Best Bias (579)
User Bias (0)

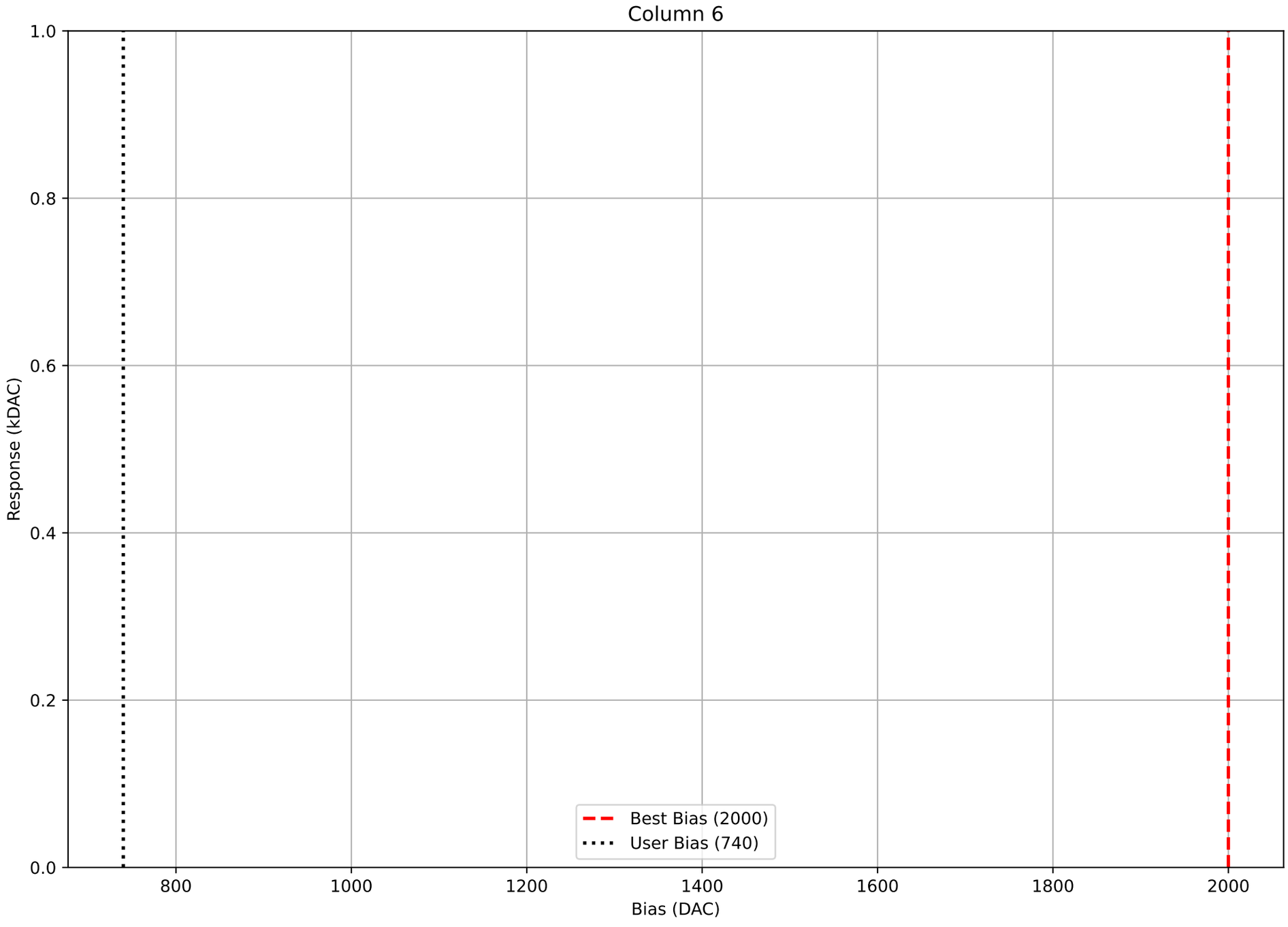

Column 1
Response (kDAC)
Bias (DAC)
20
0
−20
−40
−60
0
250
500
750
1000
1250
1500
1750
2000
c01r08
c01r09
c01r11
c01r12
c01r15
c01r21
c01r22
c01r23
c01r26
Best Bias (232)
User Bias (250)

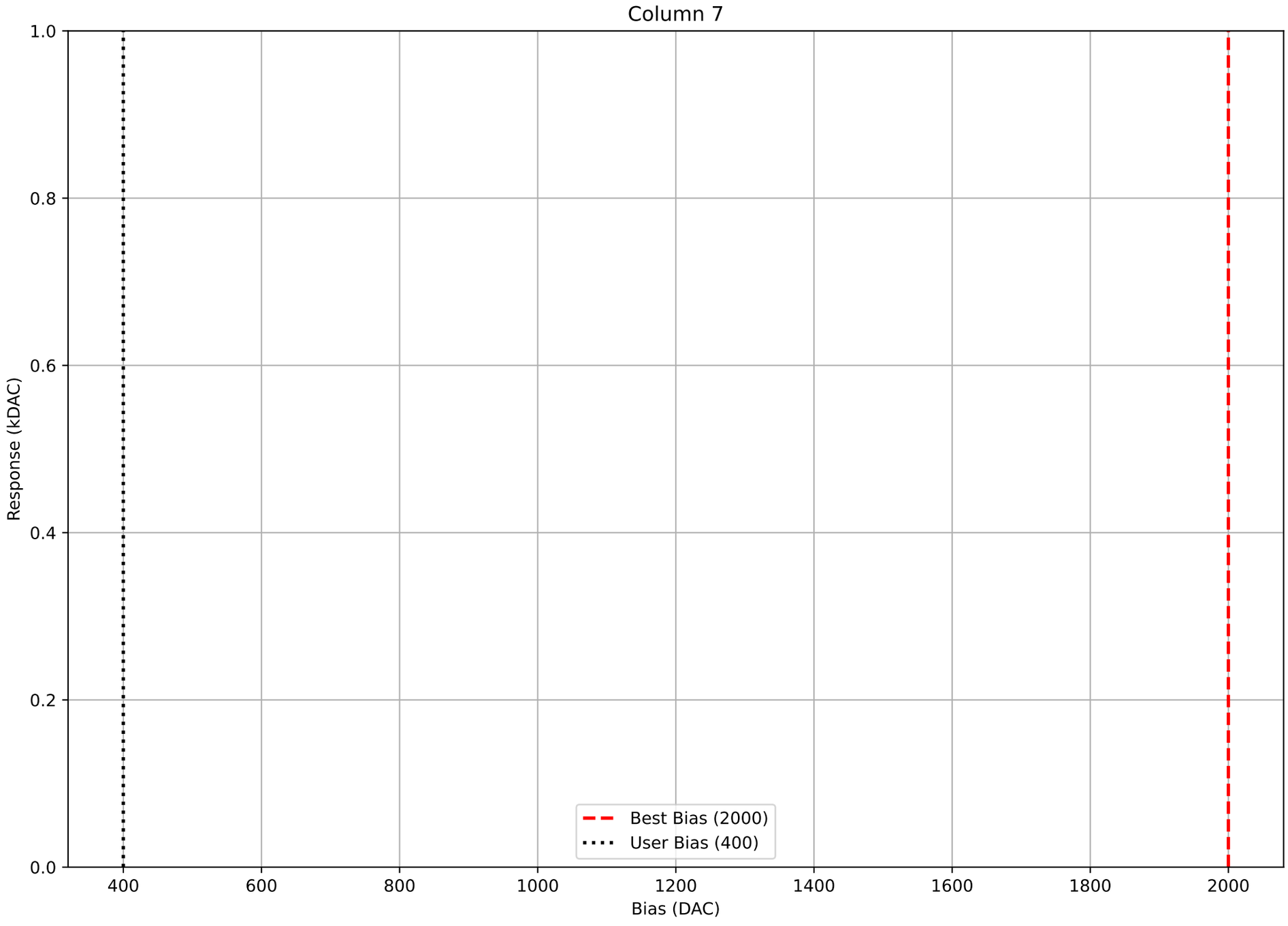

Column 2
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (700)

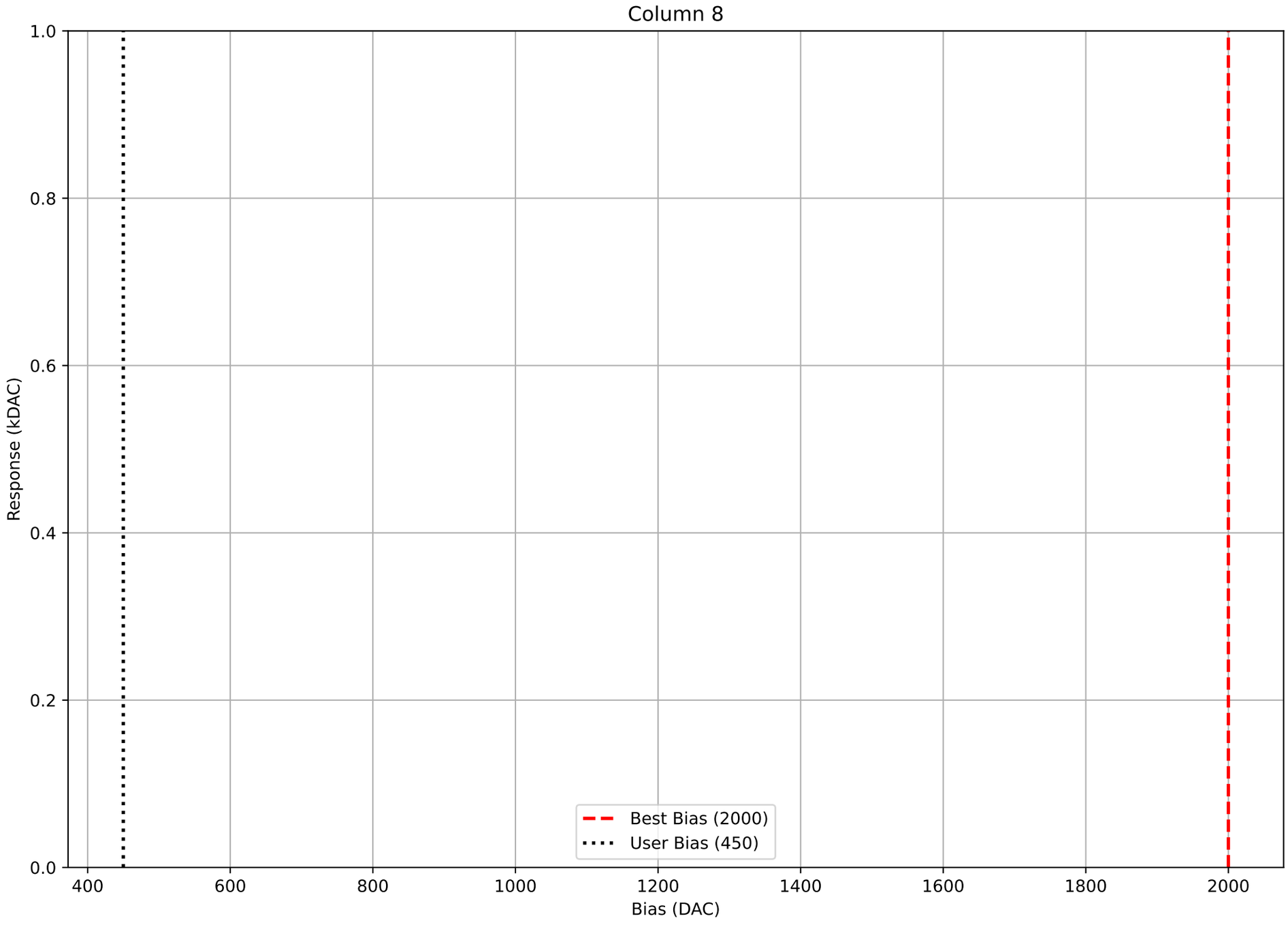

Column 3
Response (kDAC)
20
0
−20
−40
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
c03r01
c03r12
c03r16
c03r19
c03r20
c03r23
c03r24
c03r27
Best Bias (212)
User Bias (1000)

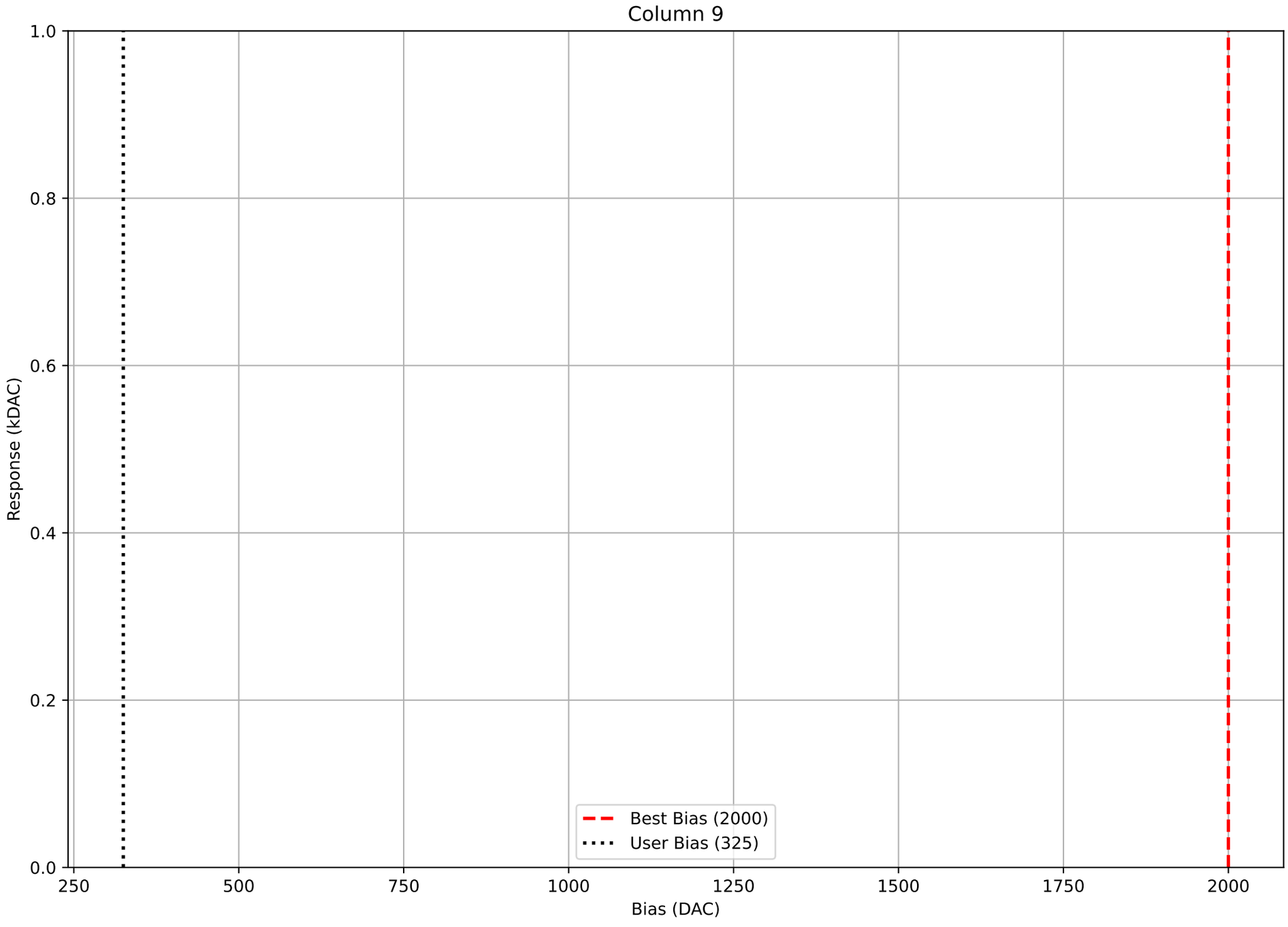

Column 4
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (1000)

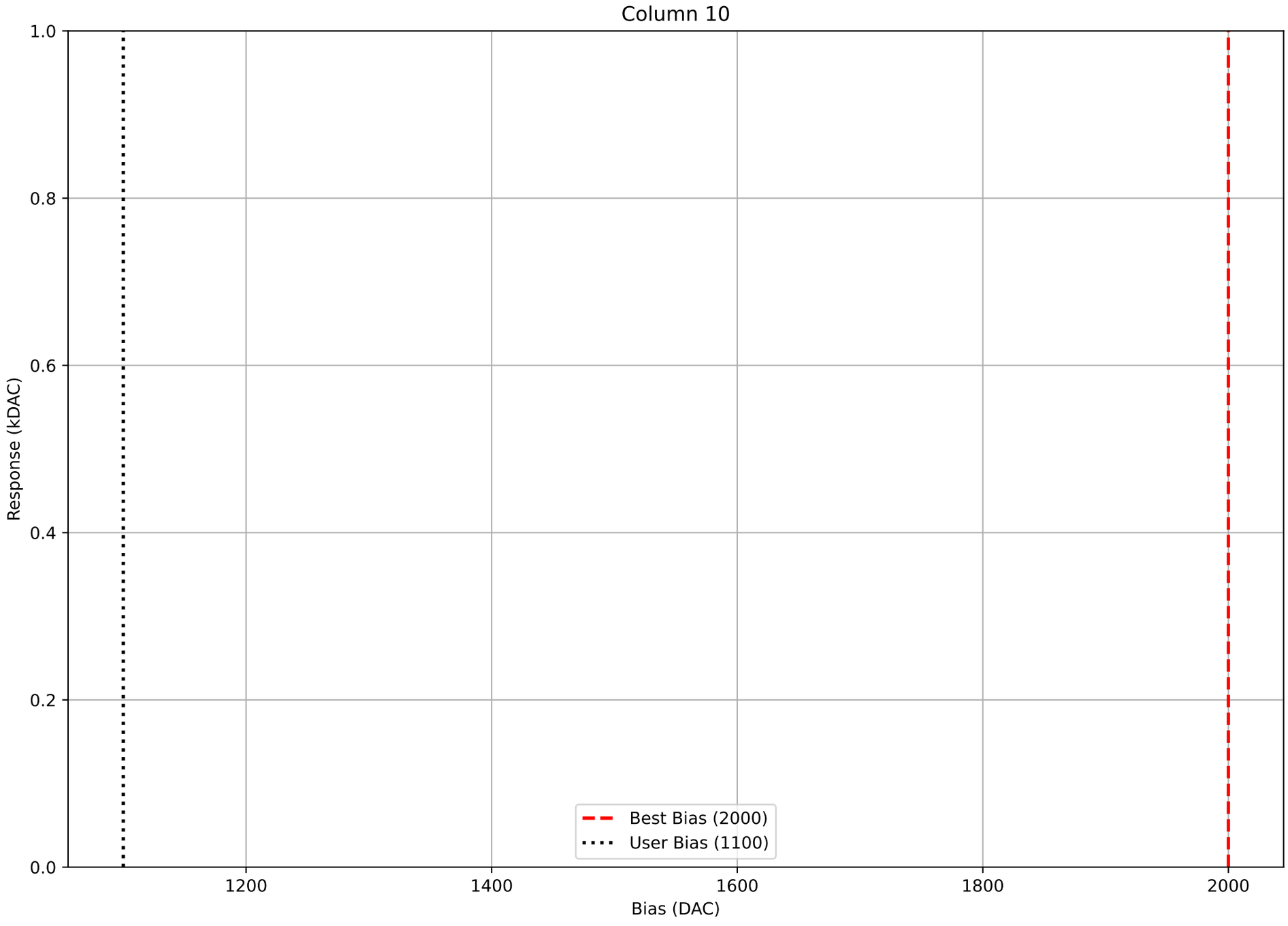

Column 5
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
1400
1500
1600
1700
1800
1900
2000
Bias (DAC)
Best Bias (2000)
User Bias (1400)

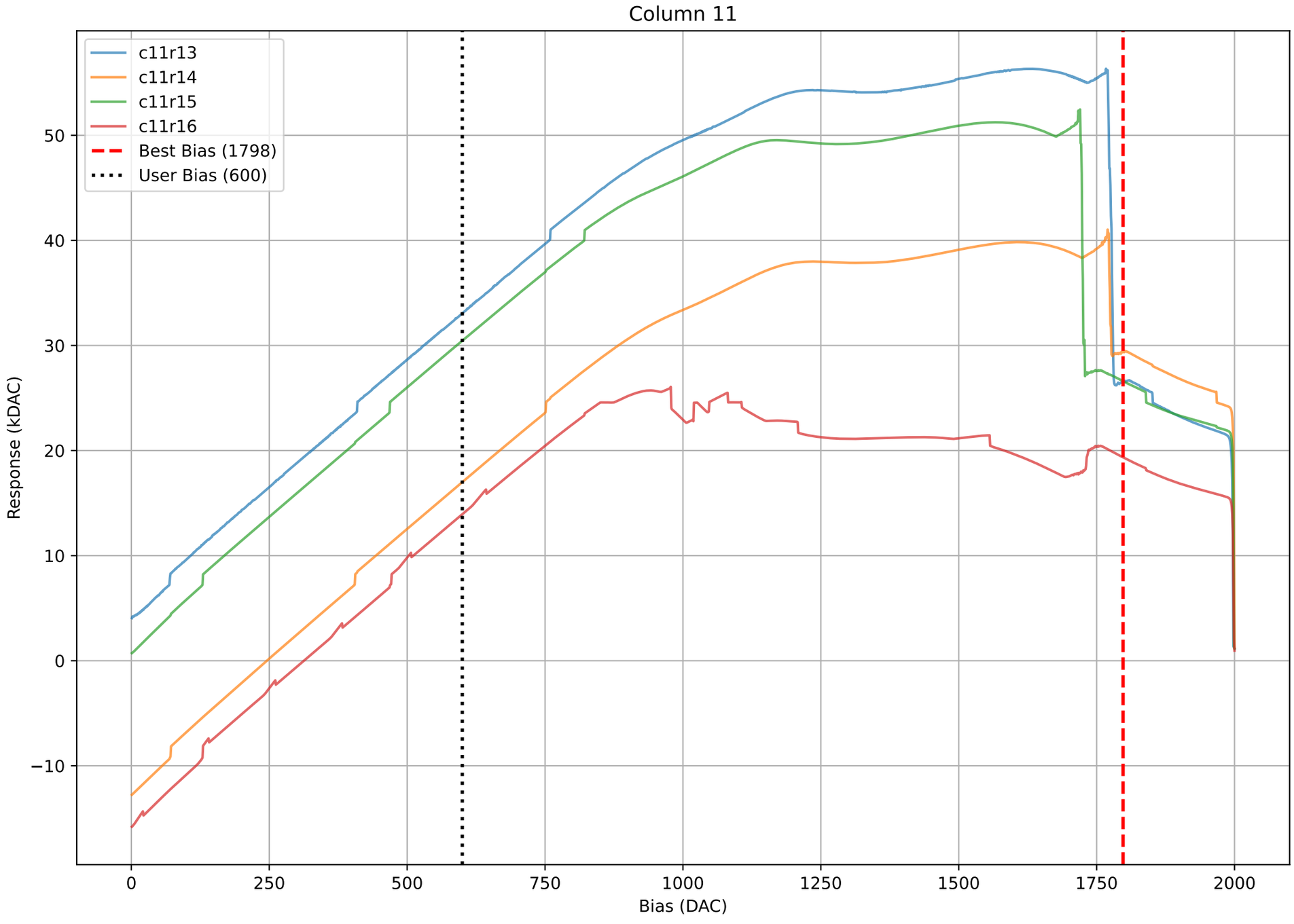

Column 6
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (740)

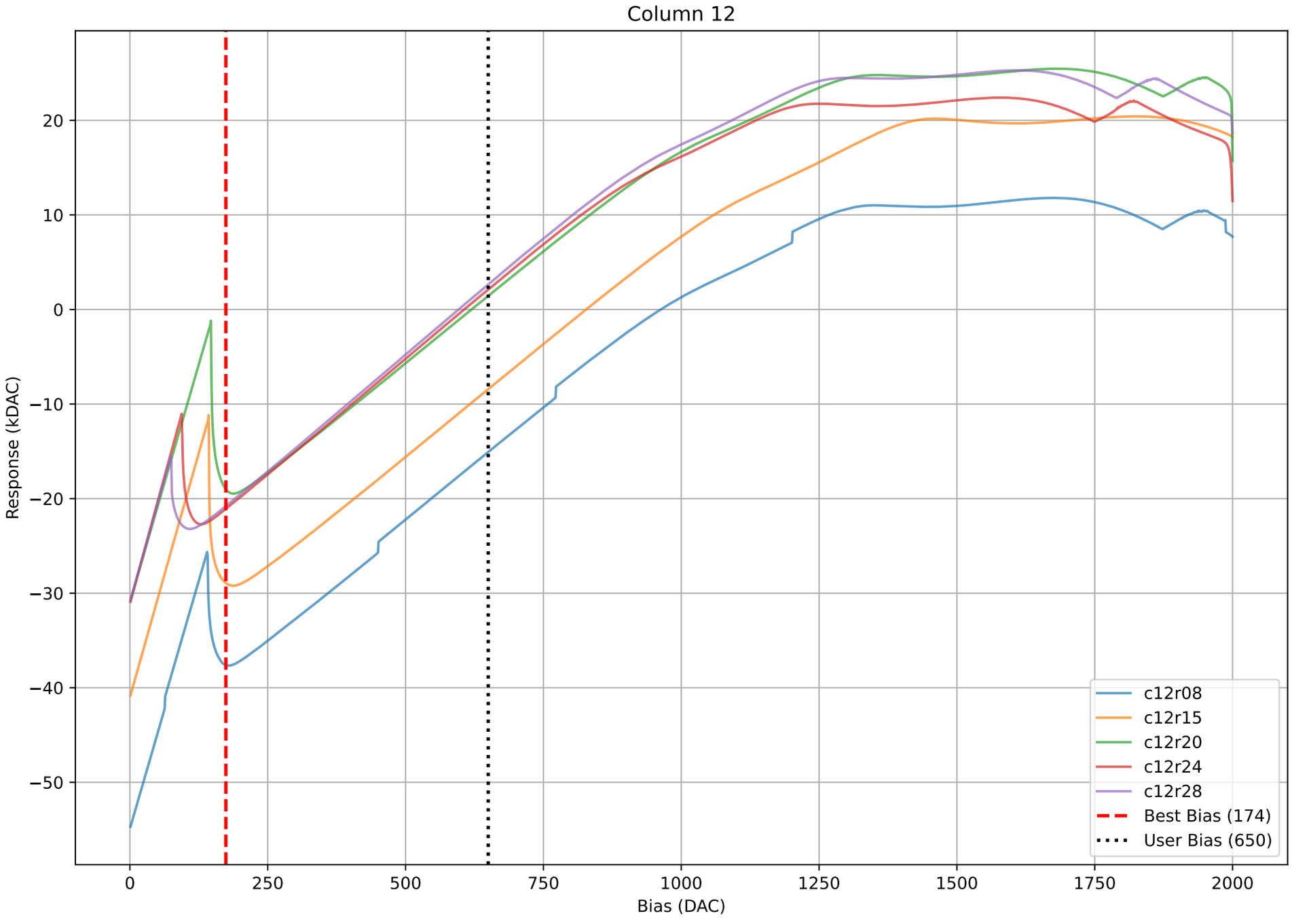

Column 7
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
400
600
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (400)

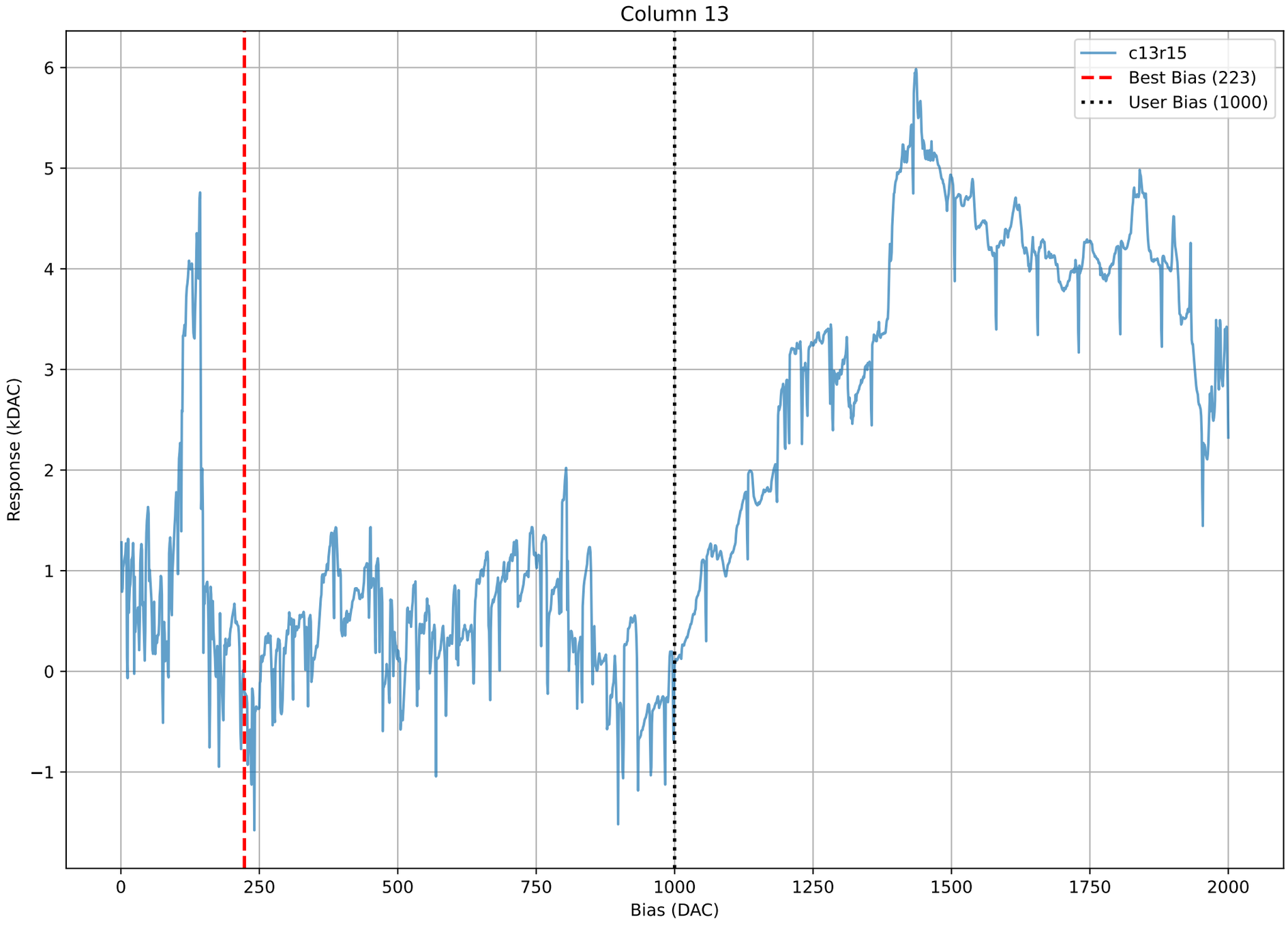

Column 8
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
400
600
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (450)

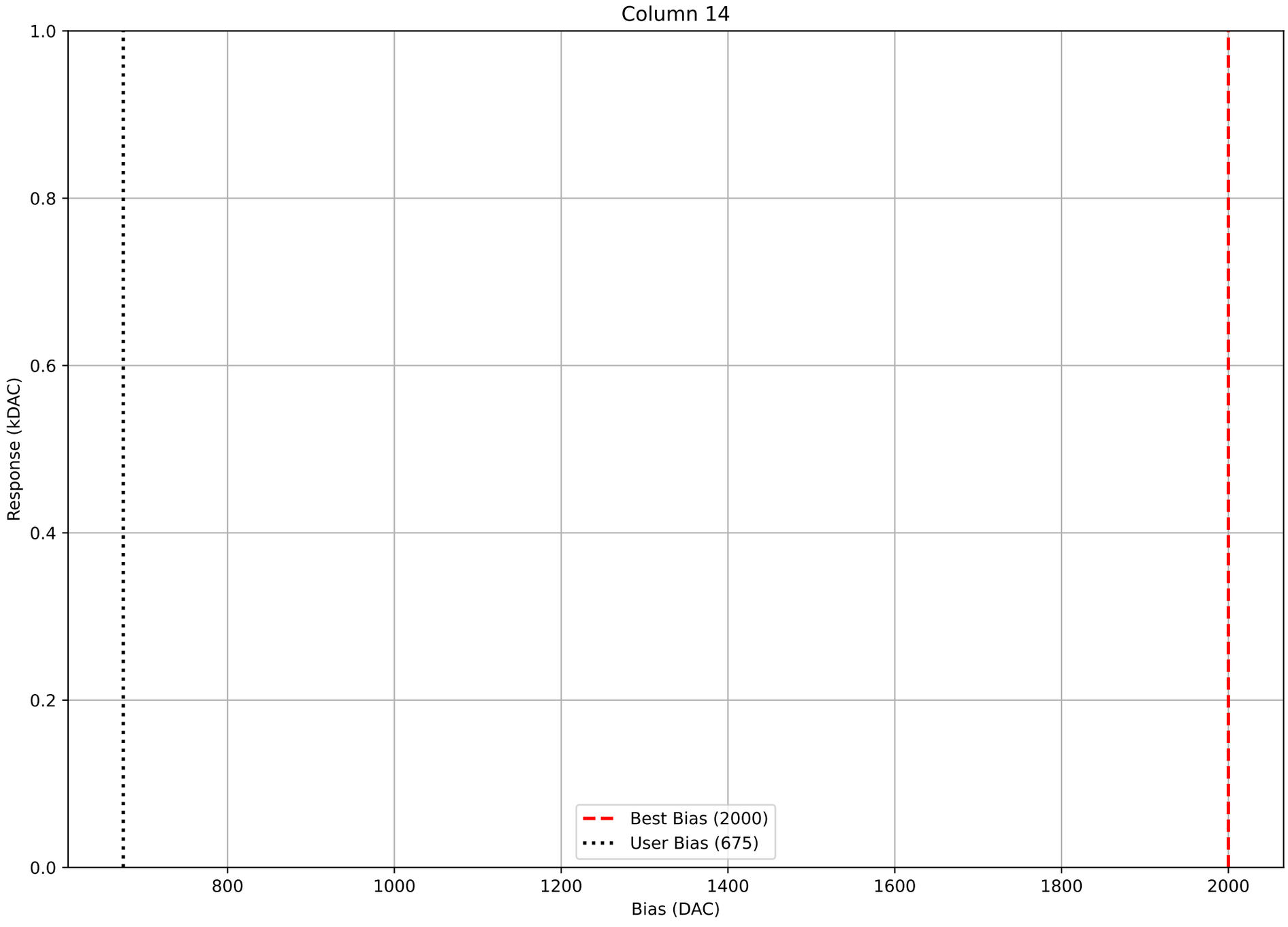

Column 9
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (325)

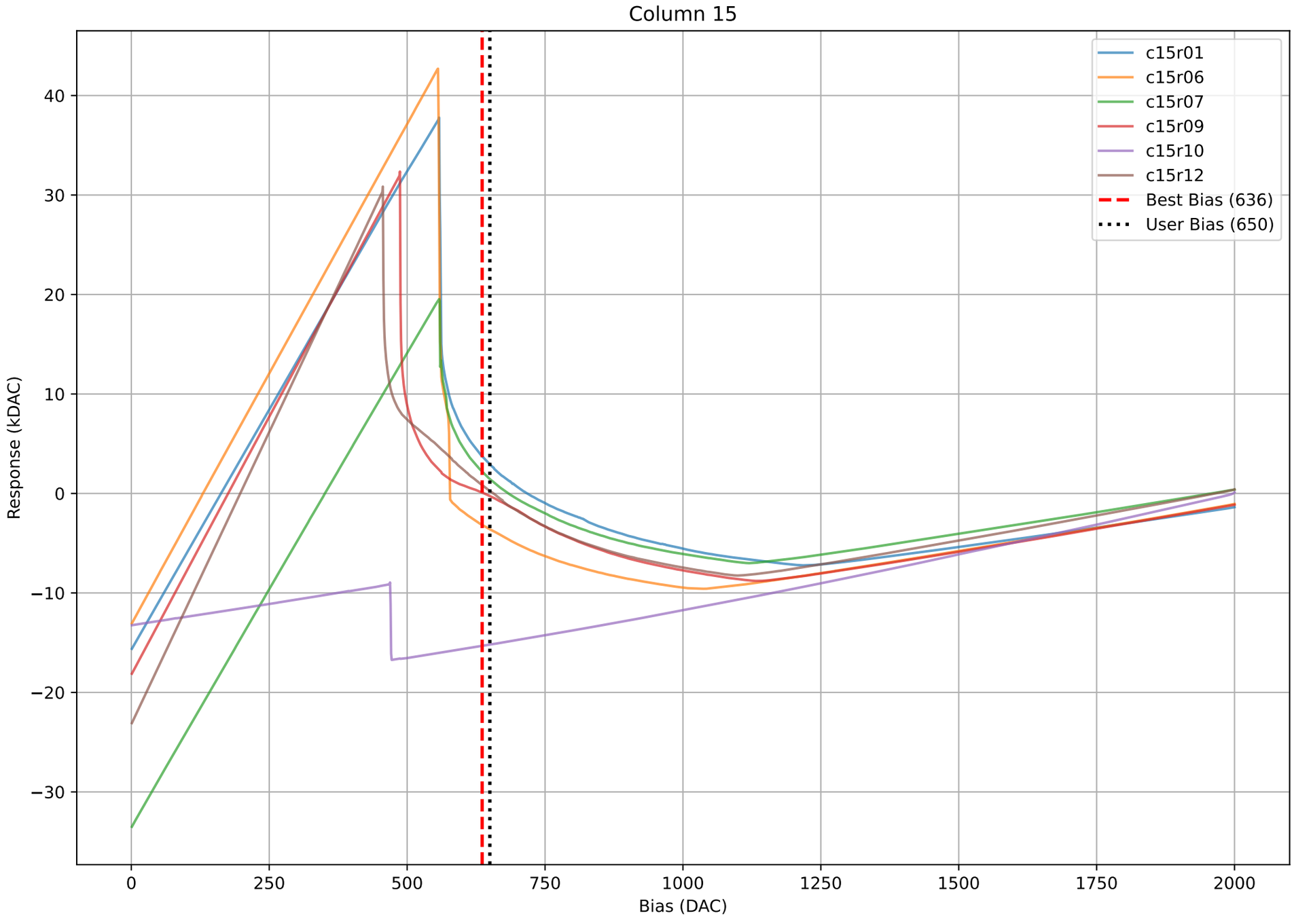

Column 10
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (1100)

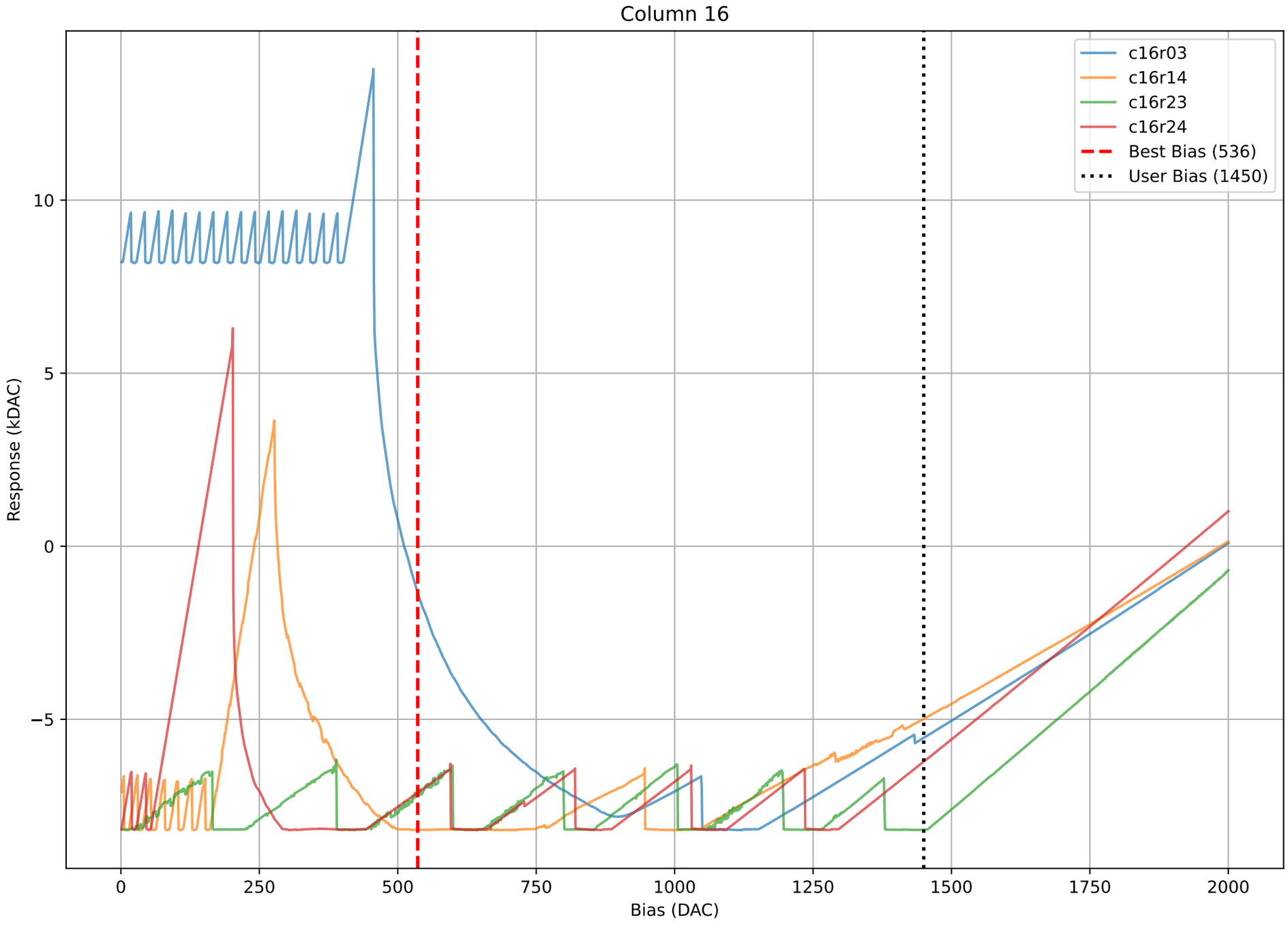

Column 11
c11r13
c11r14
c11r15
c11r16
Best Bias (1798)
User Bias (600)
Response (kDAC)
50
40
30
20
10
0
−10
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

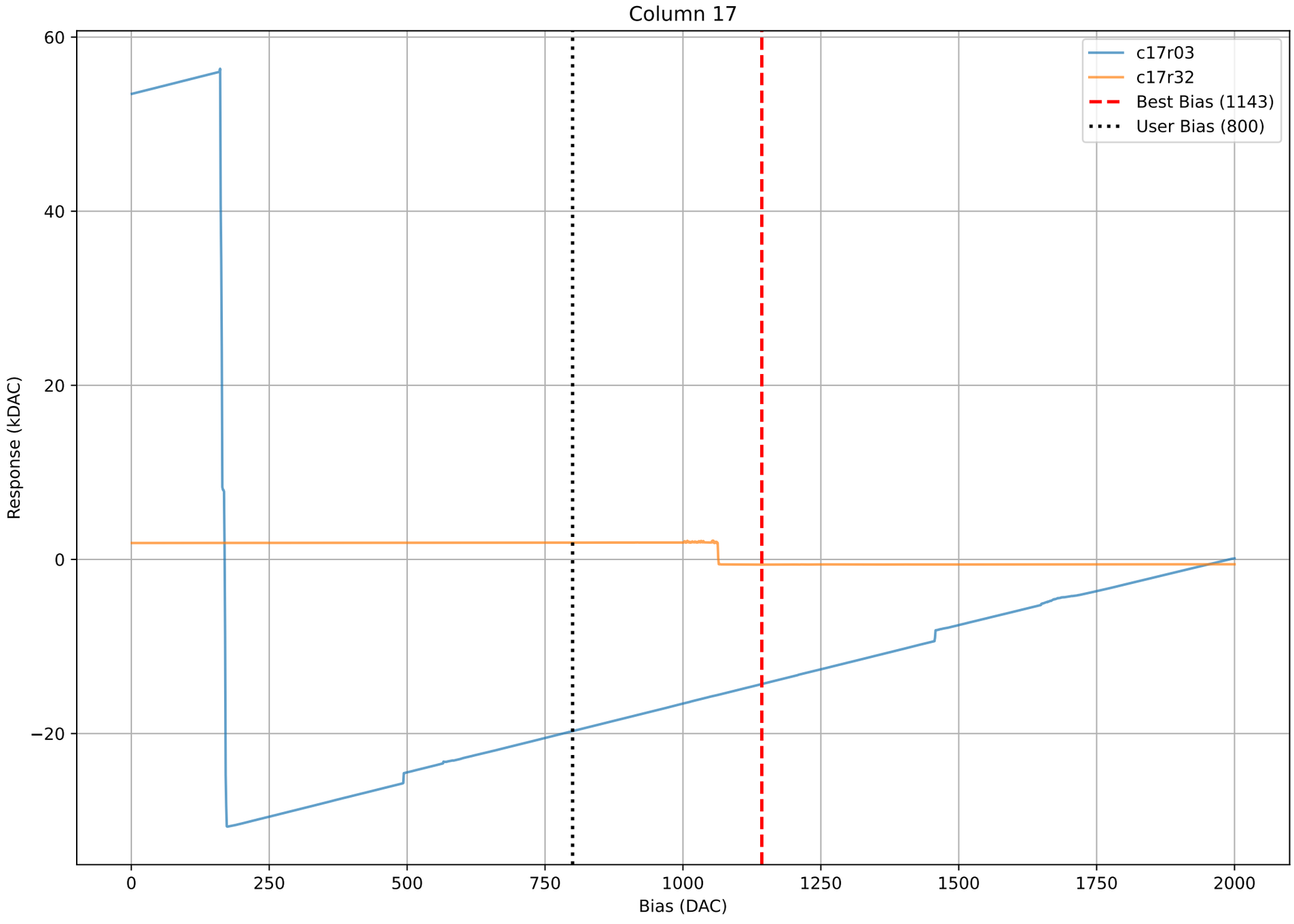

Column 12
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
c12r08
c12r15
c12r20
c12r24
c12r28
Best Bias (174)
User Bias (650)

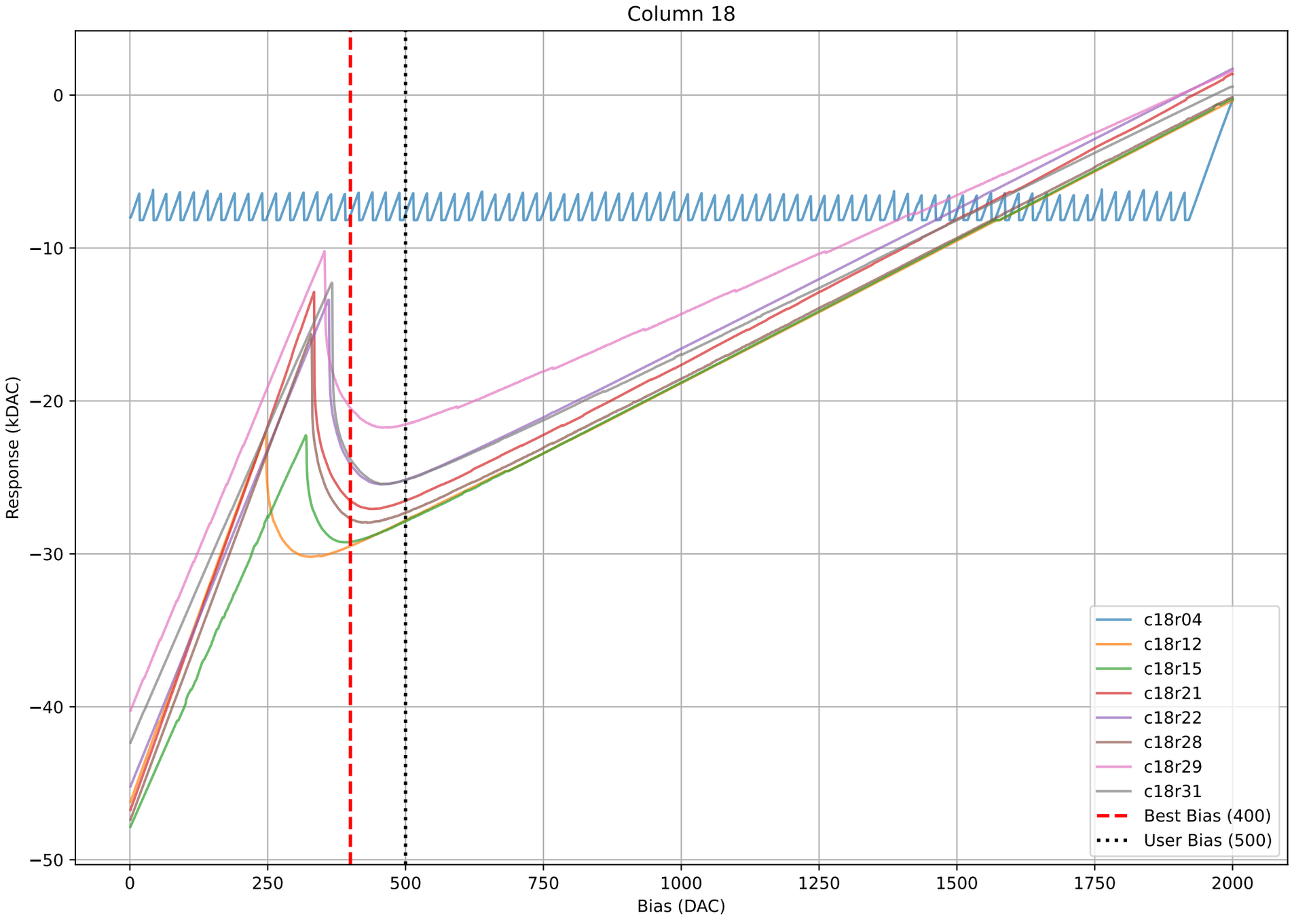

Column 13
c13r15
Best Bias (223)
User Bias (1000)
Response (kDAC)
Bias (DAC)
6
5
4
3
2
1
0
−1
0
250
500
750
1000
1250
1500
1750
2000

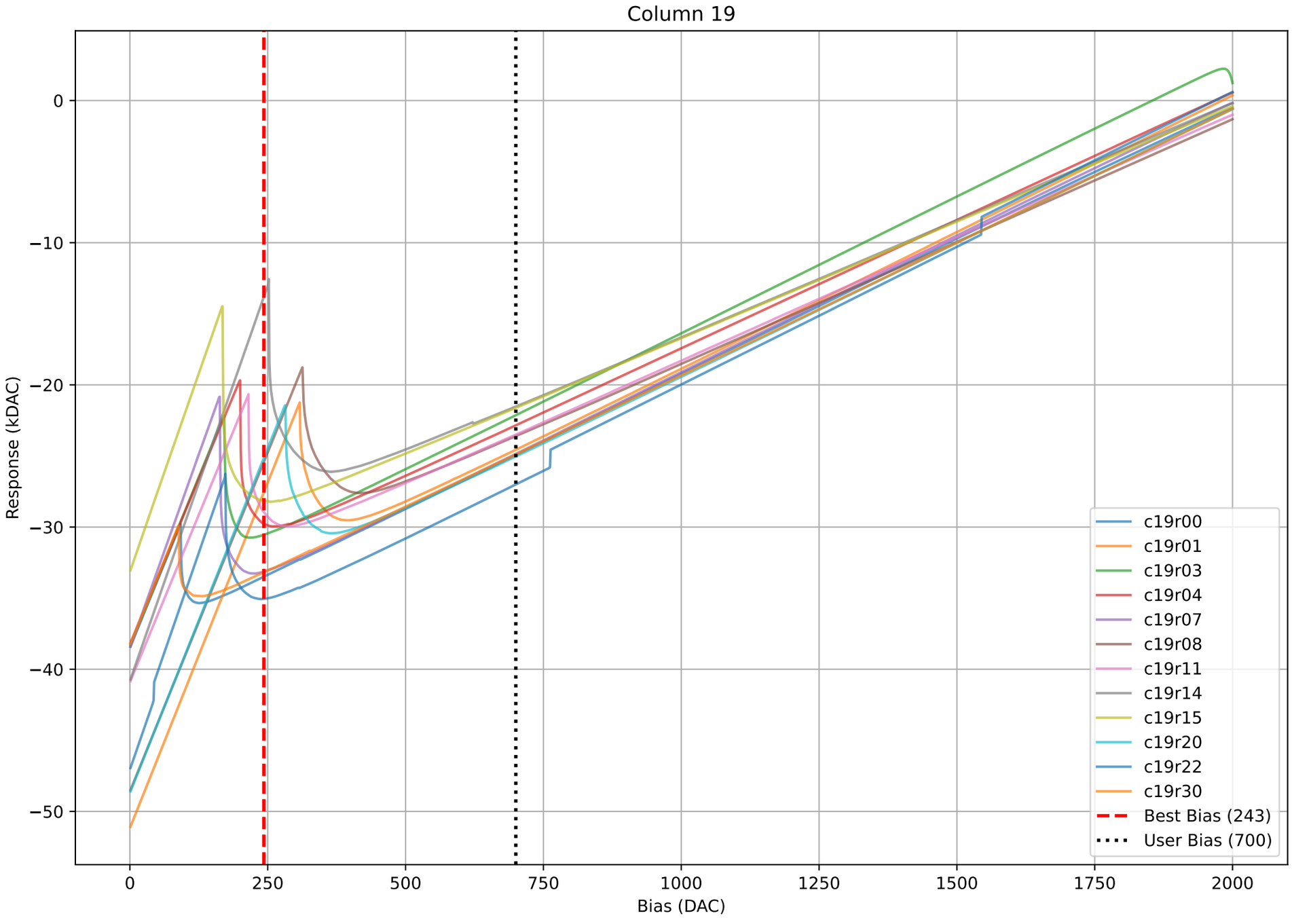

Column 14
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (675)

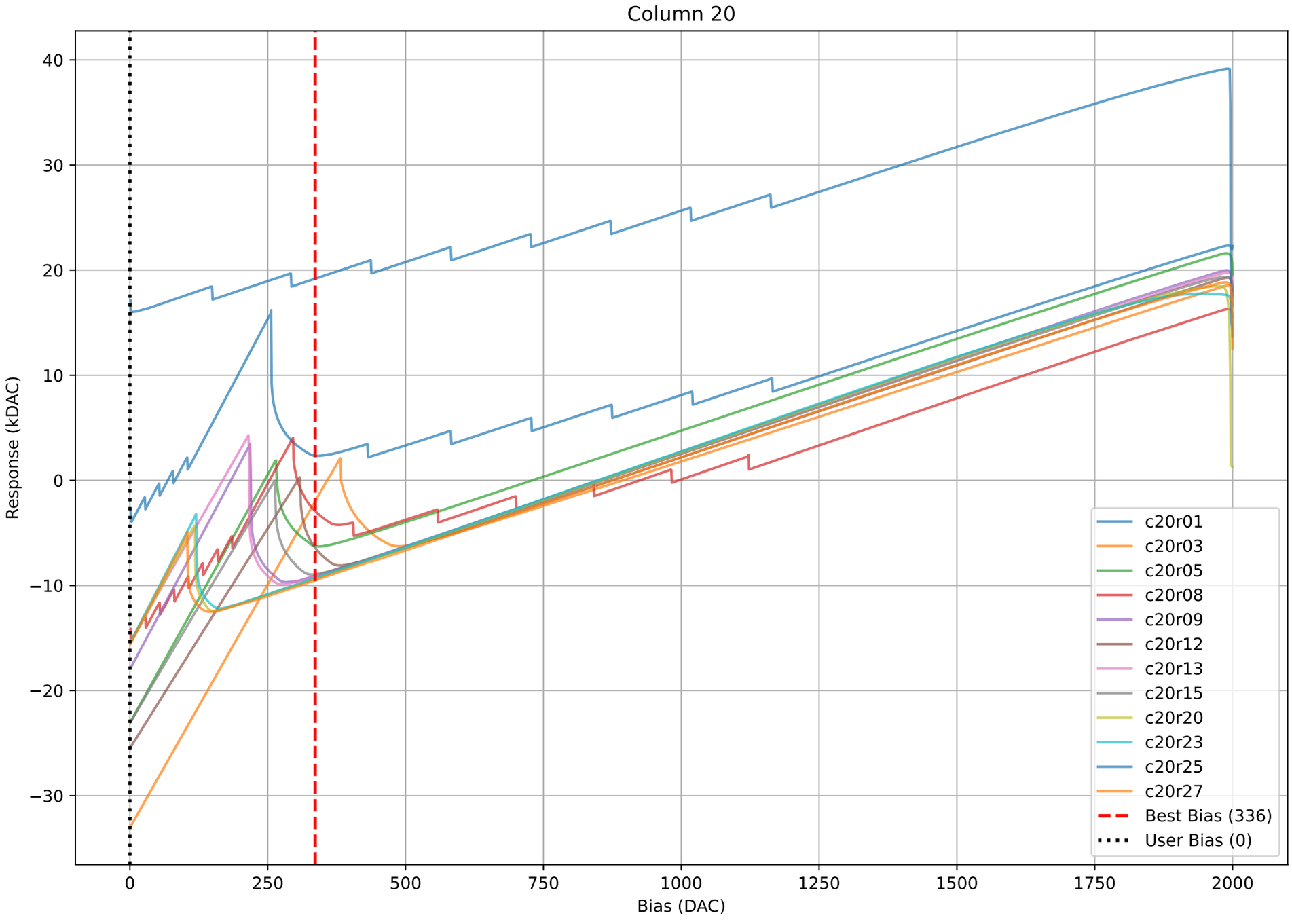

Column 15
c15r01
c15r06
c15r07
c15r09
c15r10
c15r12
Best Bias (636)
User Bias (650)
Response (kDAC)
Bias (DAC)
40
30
20
10
0
−10
−20
−30
0
250
500
750
1000
1250
1500
1750
2000

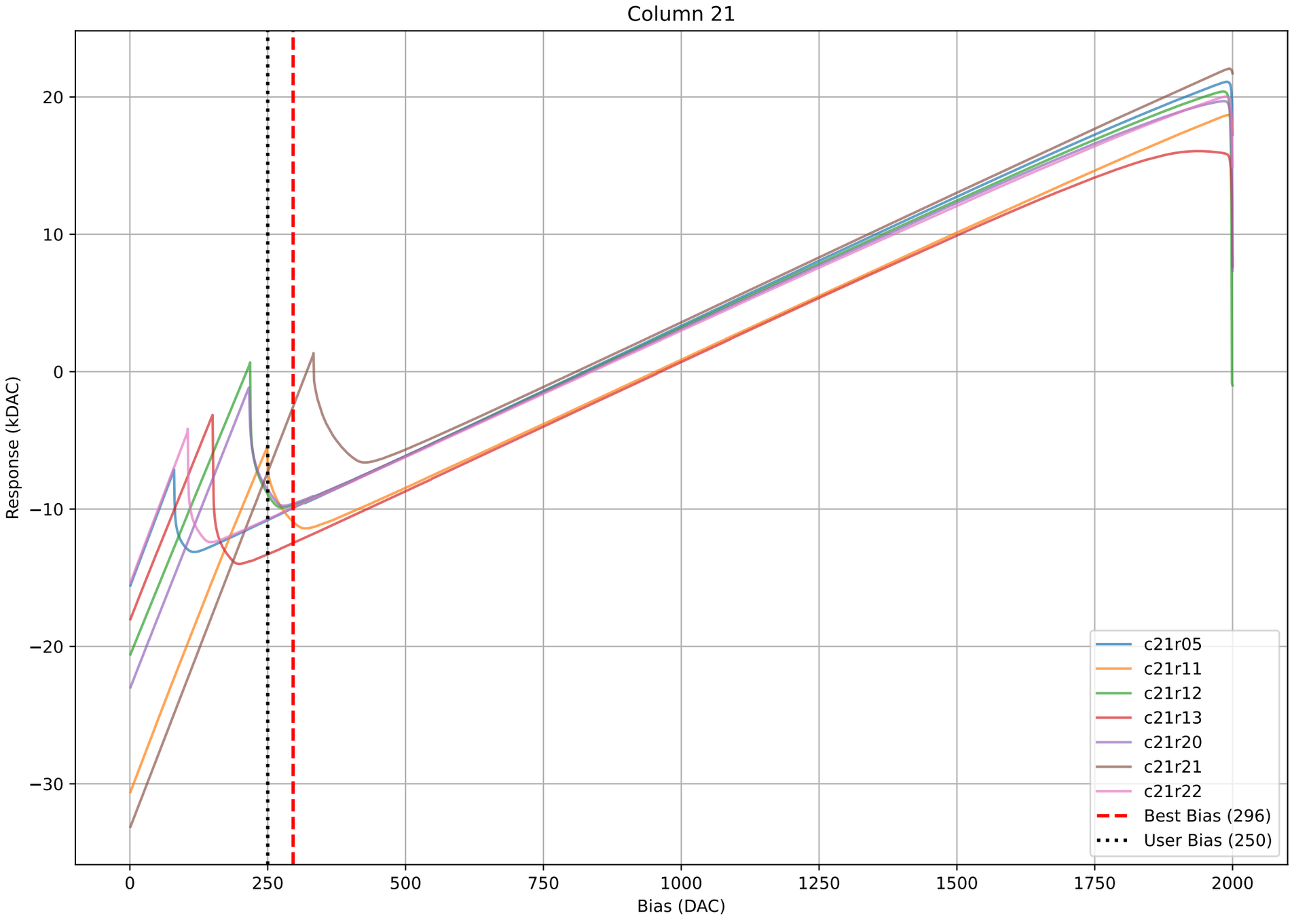

Column 16
c16r03
c16r14
c16r23
c16r24
Best Bias (536)
User Bias (1450)
Response (kDAC)
10
5
0
−5
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

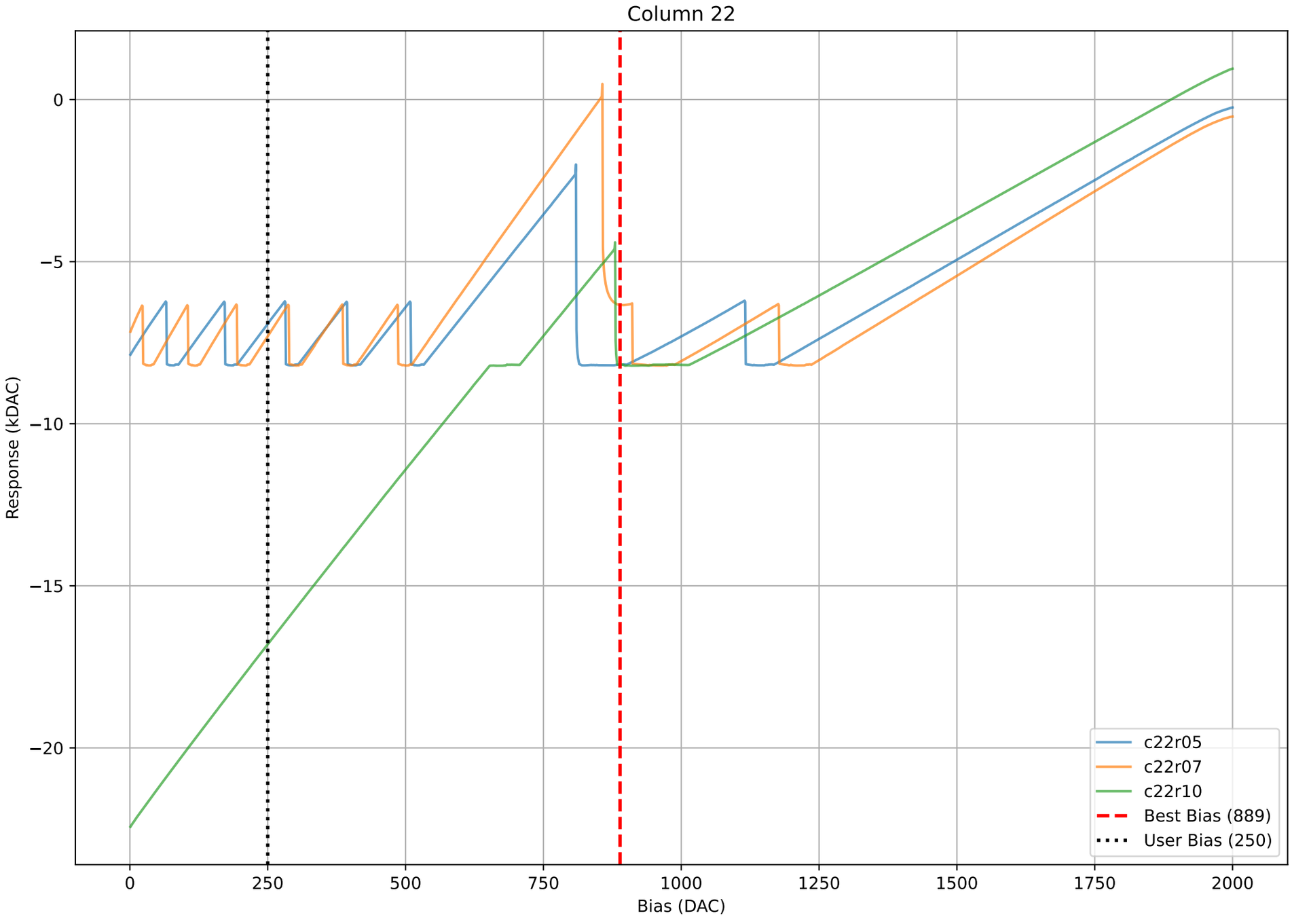

Column 17
c17r03
c17r32
Best Bias (1143)
User Bias (800)
Response (kDAC)
Bias (DAC)
60
40
20
0
−20
0
250
500
750
1000
1250
1500
1750
2000

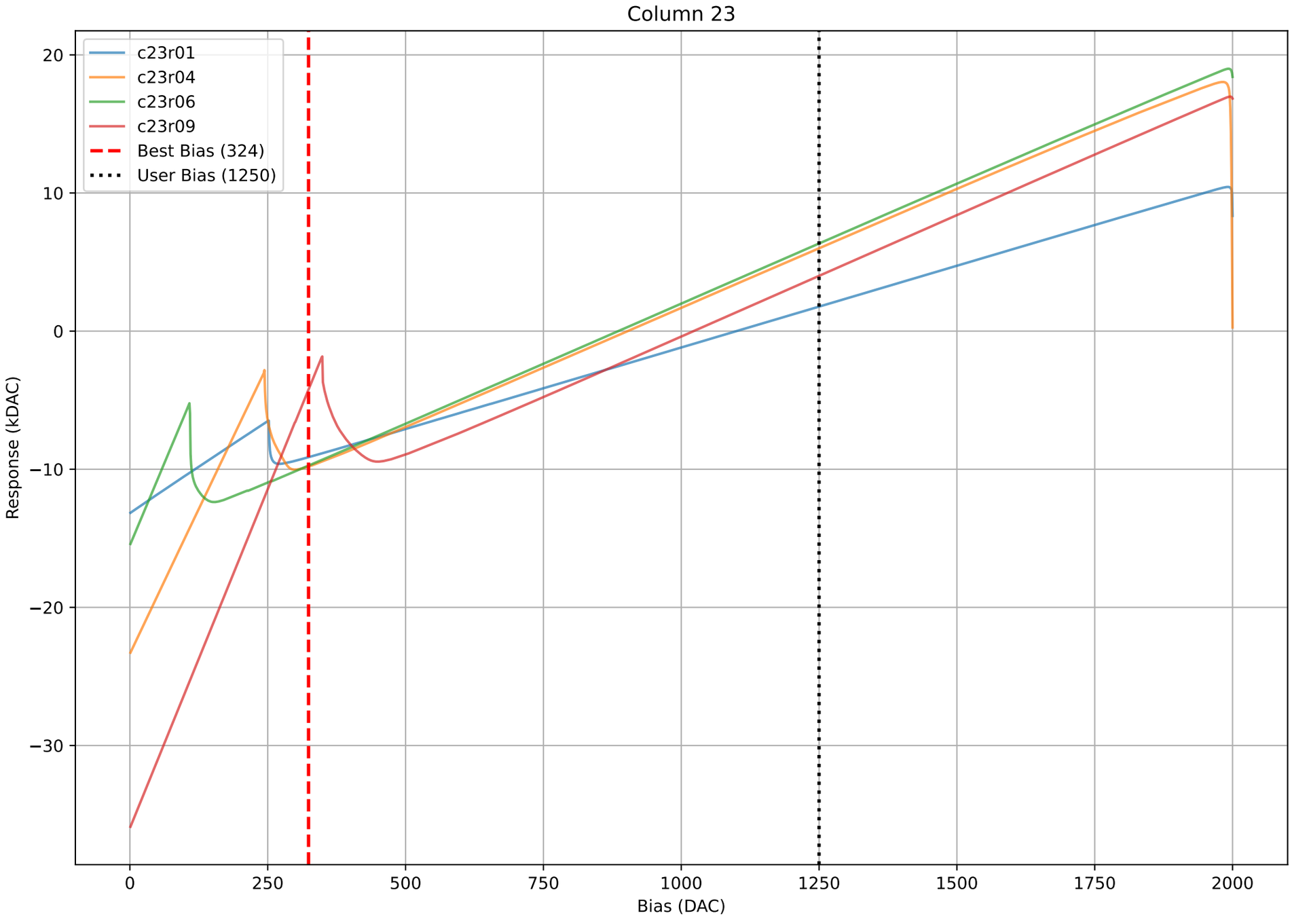

Column 18
Response (kDAC)
Bias (DAC)
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
c18r04
c18r12
c18r15
c18r21
c18r22
c18r28
c18r29
c18r31
Best Bias (400)
User Bias (500)

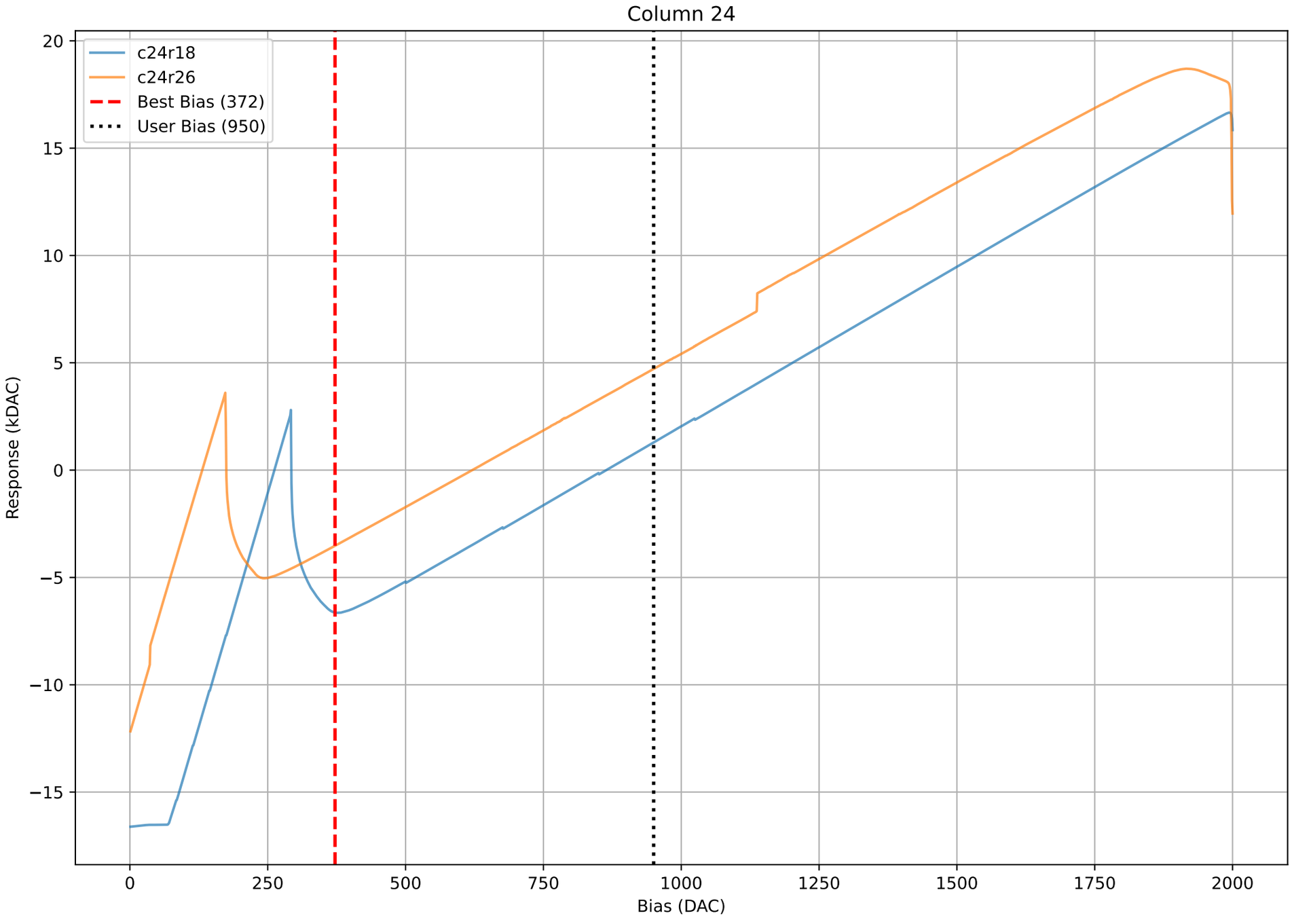

Column 19
Response (kDAC)
Bias (DAC)
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
c19r00
c19r01
c19r03
c19r04
c19r07
c19r08
c19r11
c19r14
c19r15
c19r20
c19r22
c19r30
Best Bias (243)
User Bias (700)

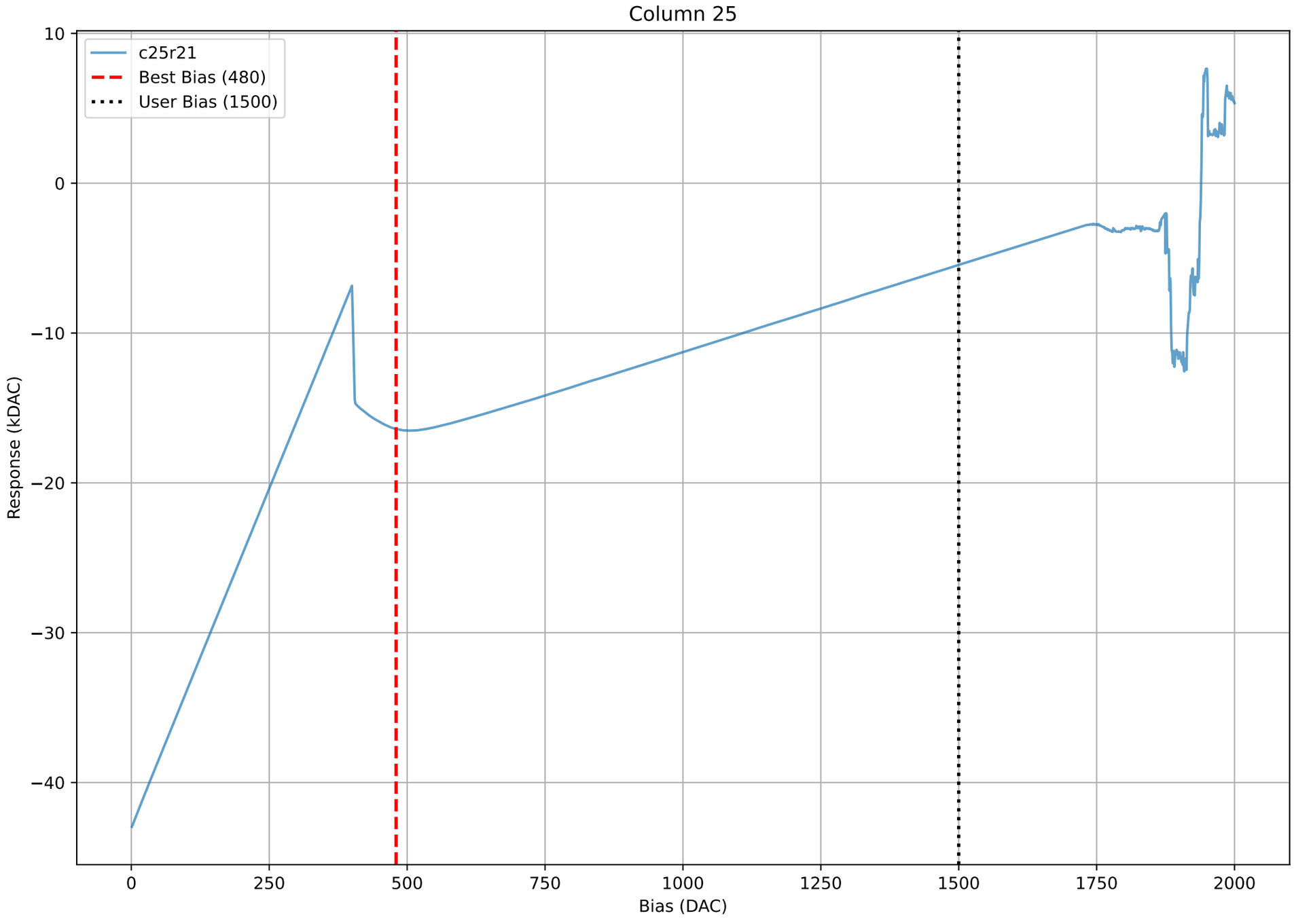

Column 20
Response (kDAC)
Bias (DAC)
40
30
20
10
0
−10
−20
−30
0
250
500
750
1000
1250
1500
1750
2000
c20r01
c20r03
c20r05
c20r08
c20r09
c20r12
c20r13
c20r15
c20r20
c20r23
c20r25
c20r27
Best Bias (336)
User Bias (0)

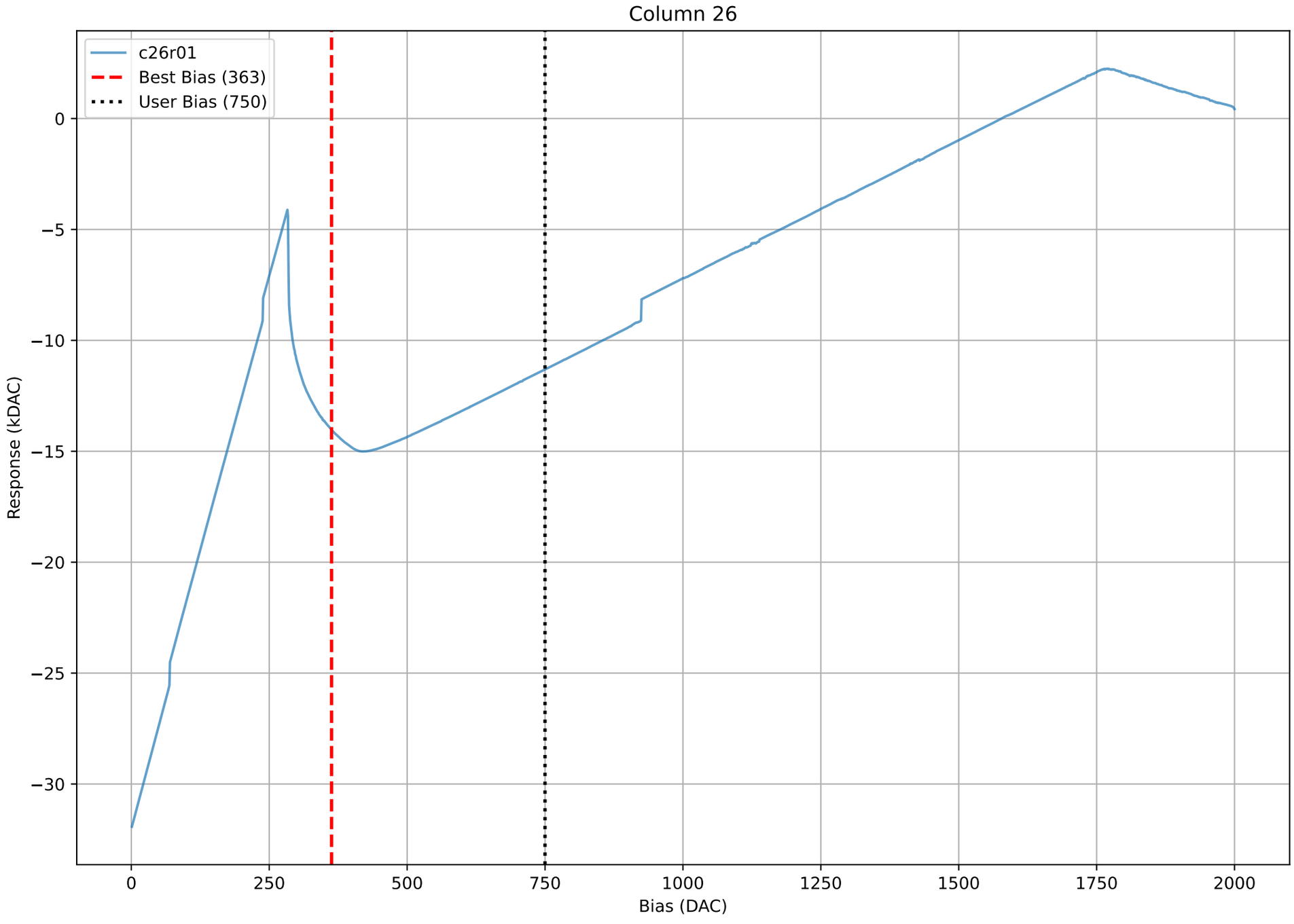

Column 21
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
−30
0
250
500
750
1000
1250
1500
1750
2000
c21r05
c21r11
c21r12
c21r13
c21r20
c21r21
c21r22
Best Bias (296)
User Bias (250)

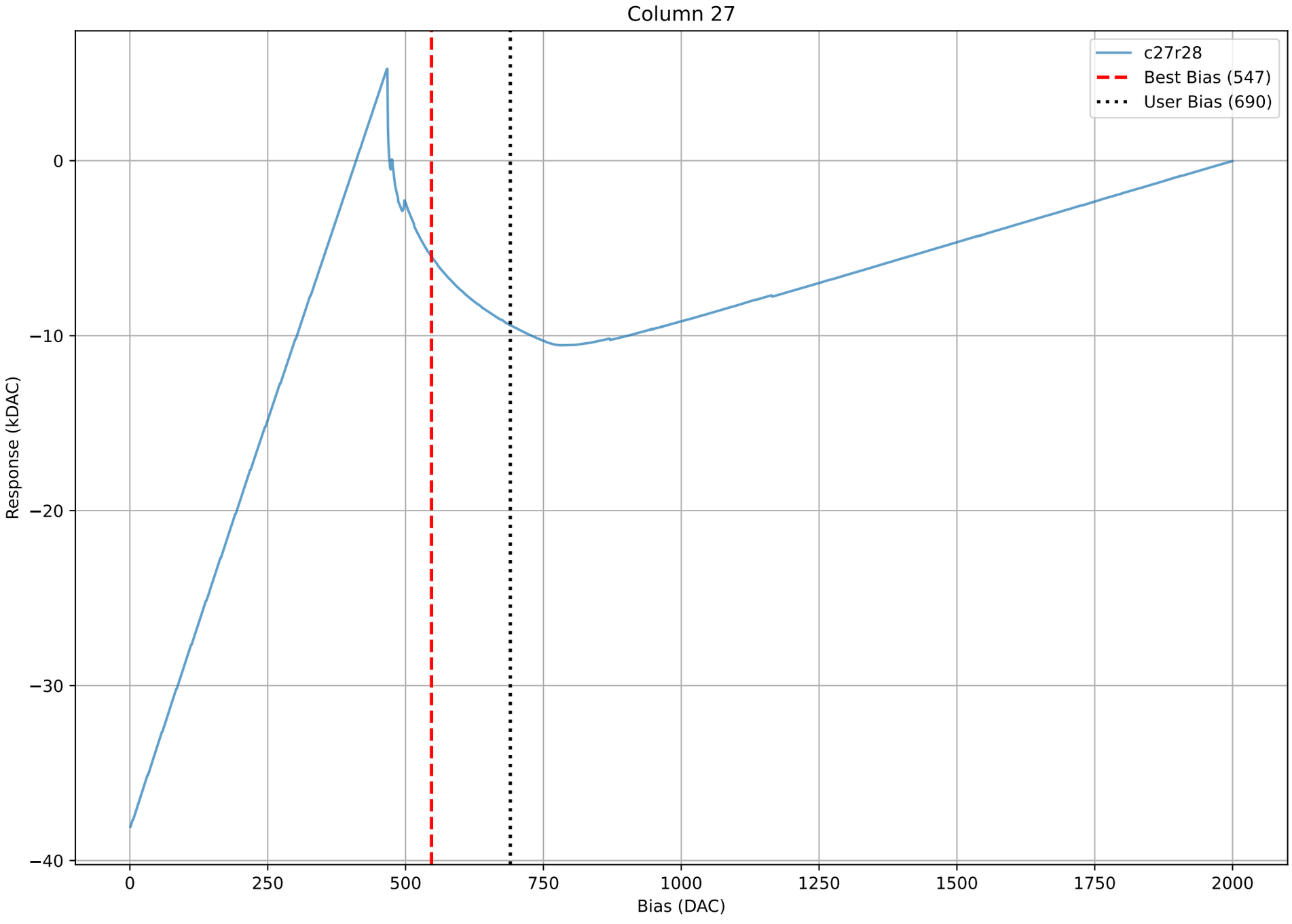

Column 22
Response (kDAC)
0
−5
−10
−15
−20
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
c22r05
c22r07
c22r10
Best Bias (889)
User Bias (250)

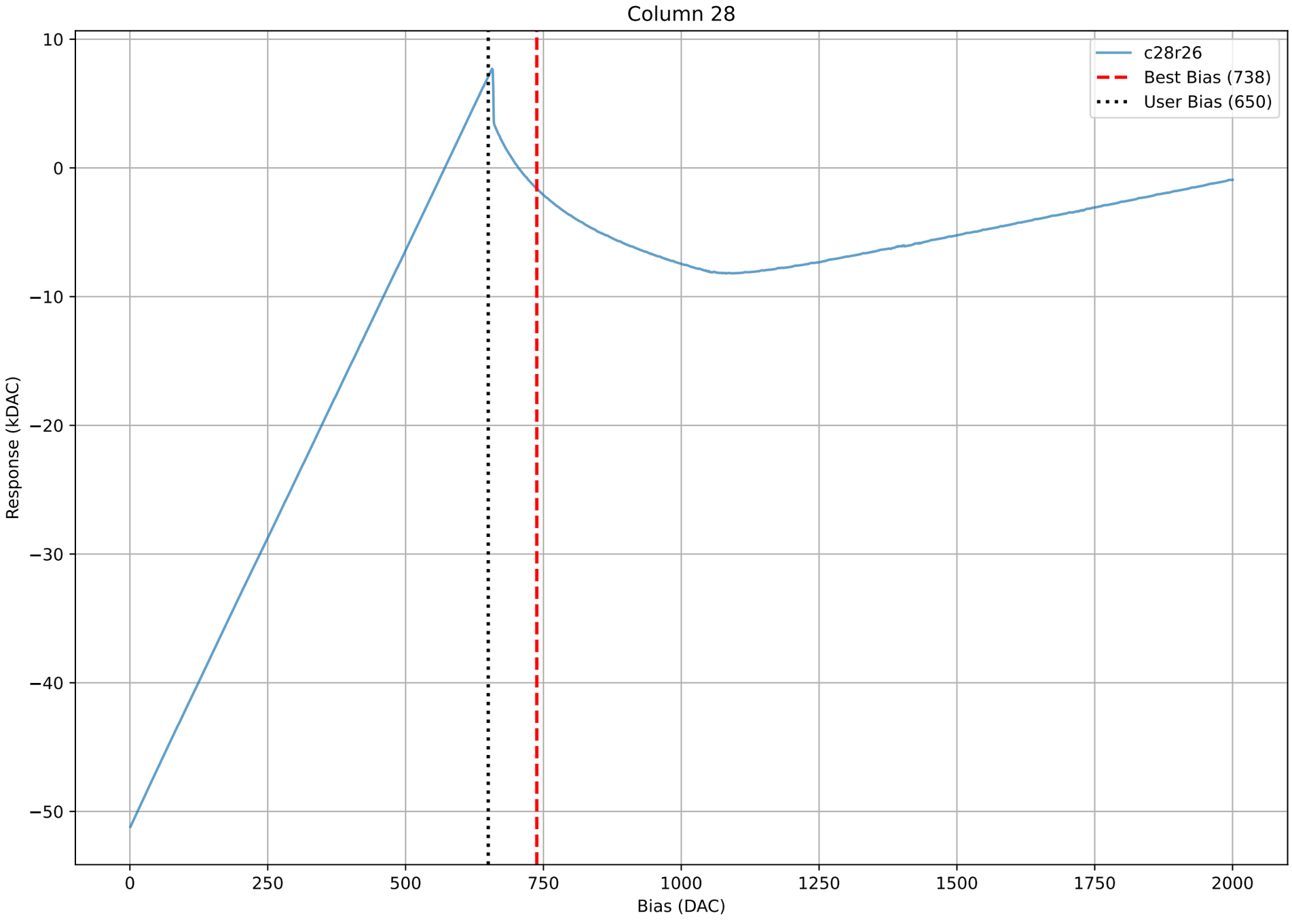

Column 23
c23r01
c23r04
c23r06
c23r09
Best Bias (324)
User Bias (1250)
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
−30
0
250
500
750
1000
1250
1500
1750
2000

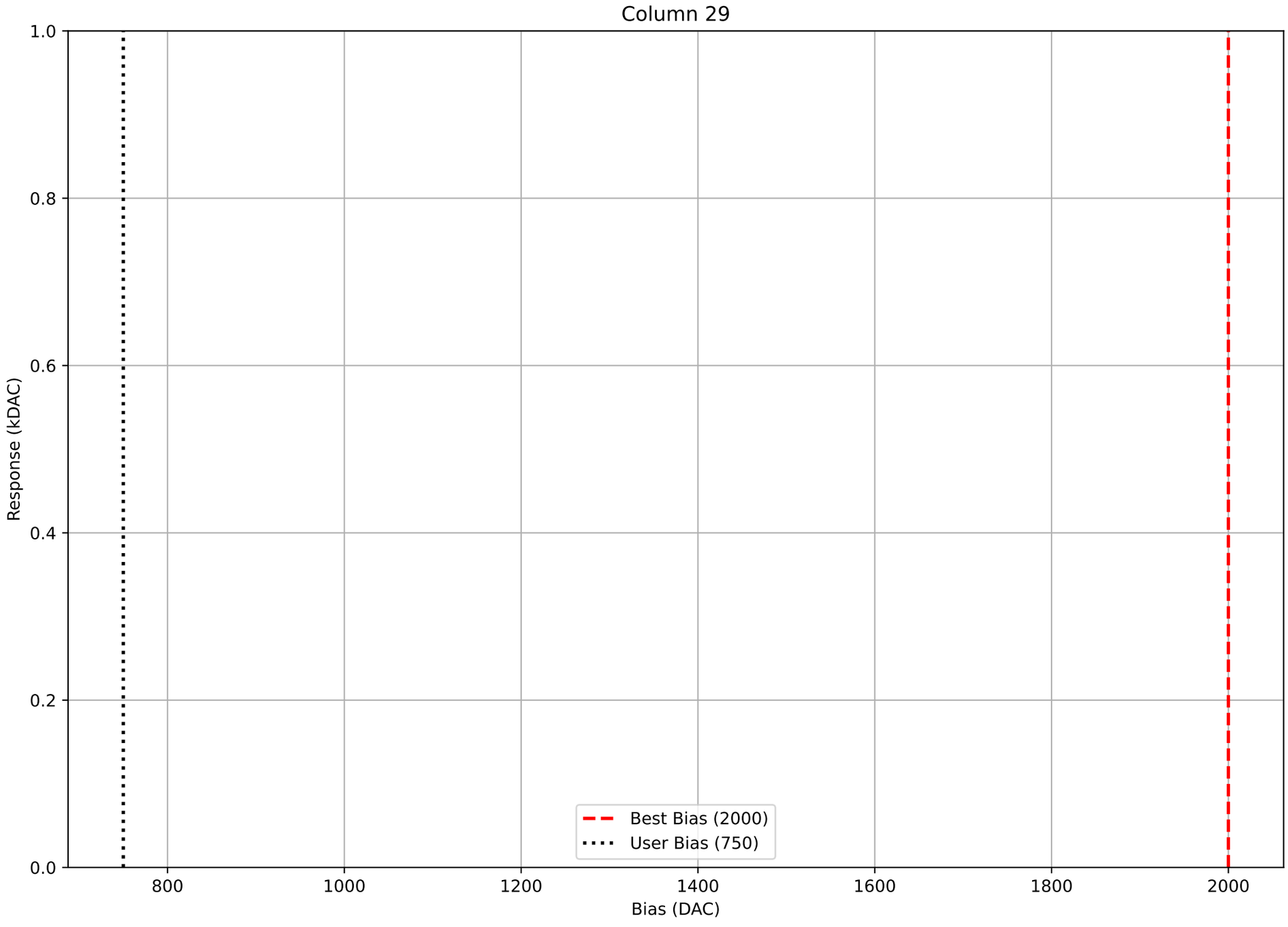

Column 24
c24r18
c24r26
Best Bias (372)
User Bias (950)
Response (kDAC)
Bias (DAC)
20
15
10
5
0
−5
−10
−15
0
250
500
750
1000
1250
1500
1750
2000

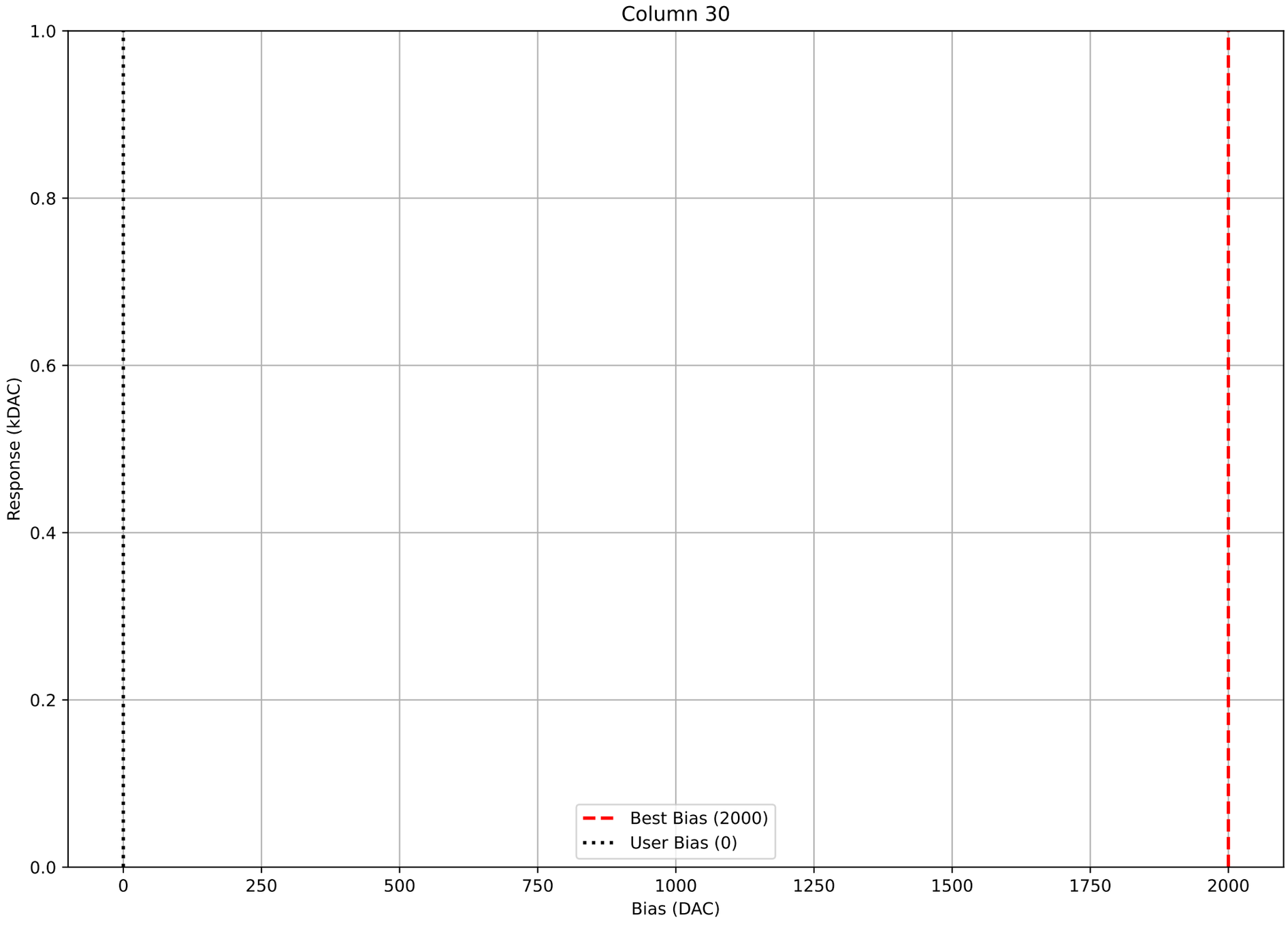

Column 25
c25r21
Best Bias (480)
User Bias (1500)
Response (kDAC)
Bias (DAC)
10
0
−10
−20
−30
−40
0
250
500
750
1000
1250
1500
1750
2000

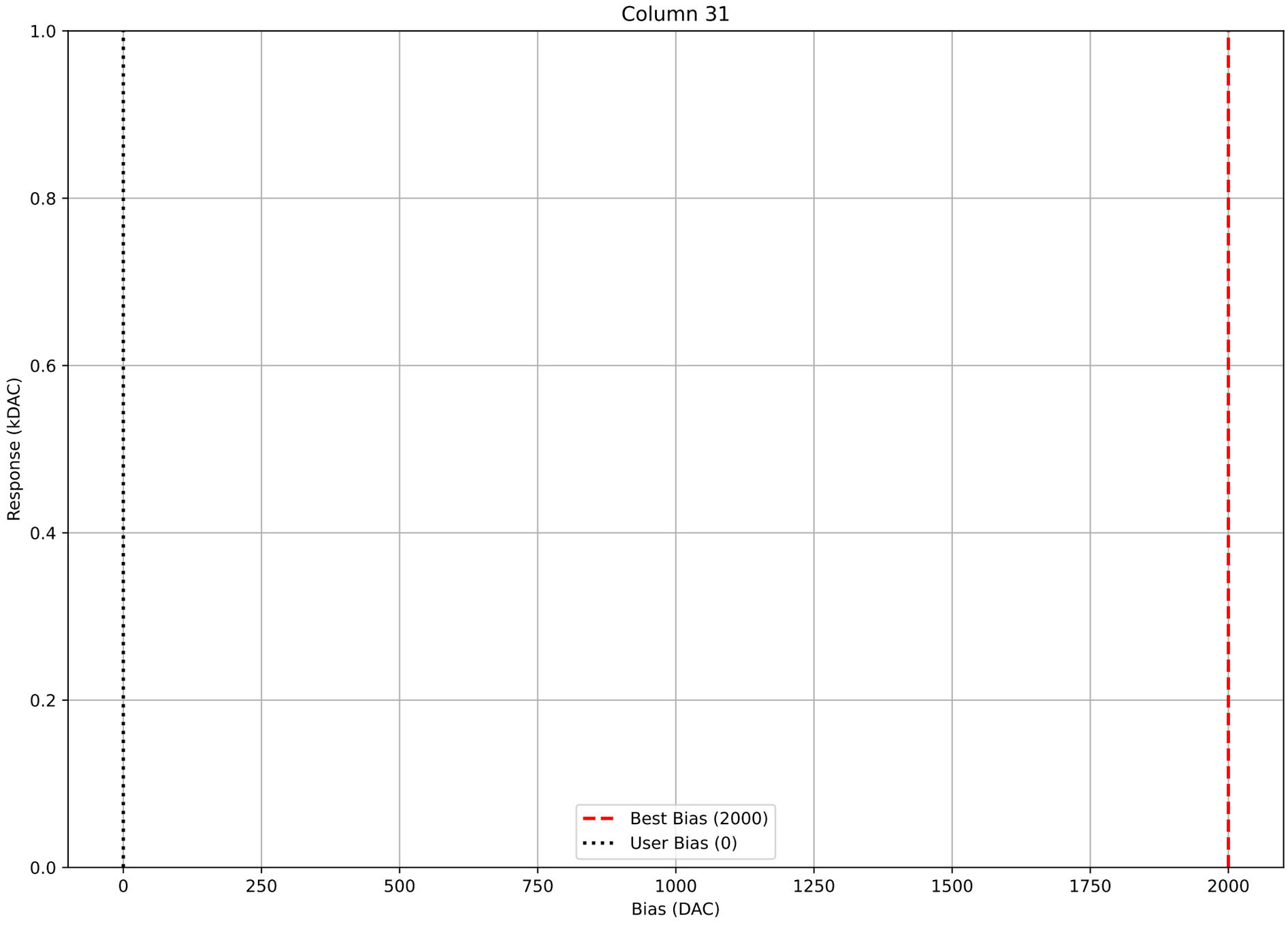

Column 26
c26r01
Best Bias (363)
User Bias (750)
Response (kDAC)
0
−5
−10
−15
−20
−25
−30
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)


Column 27
c27r28
Best Bias (547)
User Bias (690)
Response (kDAC)
0
−10
−20
−30
−40
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)


Column 28
c28r26
Best Bias (738)
User Bias (650)
Response (kDAC)
Bias (DAC)
10
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000


Column 29
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (750)


Column 30
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)


Column 31
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

```
Column 0
  Noisy detectors excluded: [(14, 0)]
    row=14 col=0 noisy_transition=[1415.0, 1411.0] min_slope=-1.196 mean_slope=-0.525 drop_kdac=-2.629
    row=14 col=0 noisy_transition=[1397.0, 1395.0] min_slope=-1.217 mean_slope=-0.827 drop_kdac=-2.434
    row=14 col=0 noisy_transition=[1187.0, 1185.0] min_slope=-1.234 mean_slope=-0.826 drop_kdac=-2.467
  Best bias = 579.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [499.0, 579.0]: [(1, 0), (16, 0)]
    row=1 col=0 transition=[519.0, 499.0] min_slope=-3.815 mean_slope=-0.574 drop_kdac=-12.217
    row=16 col=0 transition=[512.0, 500.0] min_slope=-1.023 mean_slope=-0.369 drop_kdac=-4.946
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 1
  Noisy detectors excluded: []
  Best bias = 232.0
  Max detectors in transition at best bias = 6
  Detectors at best bias in [152.0, 232.0]: [(8, 1), (9, 1), (11, 1), (12, 1), (15, 1), (23, 1)]
    row=8 col=1 transition=[167.0, 159.0] min_slope=-1.474 mean_slope=-0.463 drop_kdac=-4.166
    row=9 col=1 transition=[184.0, 172.0] min_slope=-1.371 mean_slope=-0.422 drop_kdac=-5.479
    row=11 col=1 transition=[202.0, 190.0] min_slope=-2.182 mean_slope=-0.543 drop_kdac=-7.047
    row=12 col=1 transition=[165.0, 154.0] min_slope=-1.000 mean_slope=-0.326 drop_kdac=-3.915
    row=15 col=1 transition=[192.0, 178.0] min_slope=-1.451 mean_slope=-0.388 drop_kdac=-5.833
    row=23 col=1 transition=[166.0, 152.0] min_slope=-1.058 mean_slope=-0.324 drop_kdac=-4.868
  User bias = 250
  Clean detectors in transition in [170.0, 250.0]: [(9, 1), (11, 1), (15, 1)]
    row=9 col=1 transition=[184.0, 172.0] min_slope=-1.371 mean_slope=-0.422 drop_kdac=-5.479
    row=11 col=1 transition=[202.0, 190.0] min_slope=-2.182 mean_slope=-0.543 drop_kdac=-7.047
    row=15 col=1 transition=[192.0, 178.0] min_slope=-1.451 mean_slope=-0.388 drop_kdac=-5.833
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=8 col=1 transition=[167.0, 159.0] min_slope=-1.474 mean_slope=-0.463 drop_kdac=-4.166
gap_to_window=11.0 reason=accepted transition outside user window
    row=12 col=1 transition=[165.0, 154.0] min_slope=-1.000 mean_slope=-0.326 drop_kdac=-3.915
gap_to_window=16.0 reason=accepted transition outside user window
    row=23 col=1 transition=[166.0, 152.0] min_slope=-1.058 mean_slope=-0.324 drop_kdac=-4.868
gap_to_window=18.0 reason=accepted transition outside user window

Column 2
  Noisy detectors excluded: [(14, 2), (22, 2)]
    row=14 col=2 noisy_transition=[589.0, 586.0] min_slope=-1.335 mean_slope=-0.677 drop_kdac=-2.748
    row=14 col=2 noisy_transition=[449.0, 448.0] min_slope=-1.352 mean_slope=-1.314 drop_kdac=-2.669
    row=14 col=2 noisy_transition=[267.0, 266.0] min_slope=-1.480 mean_slope=-1.433 drop_kdac=-2.944
```

```
    row=22 col=2 noisy_transition=[147.0, 146.0] min_slope=-1.274 mean_slope=-1.257 drop_kdac=-2.465
    row=22 col=2 noisy_transition=[139.0, 137.0] min_slope=-1.314 mean_slope=-0.870 drop_kdac=-2.627
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 700
  Clean detectors in transition in [620.0, 700.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=14 col=2 transition=[589.0, 586.0] min_slope=-1.335 mean_slope=-0.677 drop_kdac=-2.748
gap_to_window=34.0 reason=excluded as noisy (multiple accepted transitions)

Column 3
  Noisy detectors excluded: []
  Best bias = 212.0
  Max detectors in transition at best bias = 5
  Detectors at best bias in [132.0, 212.0]: [(16, 3), (19, 3), (20, 3), (23, 3), (27, 3)]
    row=16 col=3 transition=[145.0, 132.0] min_slope=-1.058 mean_slope=-0.326 drop_kdac=-4.569
    row=19 col=3 transition=[196.0, 177.0] min_slope=-2.833 mean_slope=-0.532 drop_kdac=-10.652
    row=20 col=3 transition=[158.0, 143.0] min_slope=-1.222 mean_slope=-0.413 drop_kdac=-6.627
    row=23 col=3 transition=[181.0, 166.0] min_slope=-2.354 mean_slope=-0.485 drop_kdac=-7.755
    row=27 col=3 transition=[145.0, 132.0] min_slope=-1.061 mean_slope=-0.326 drop_kdac=-4.566
  User bias = 1000
  Clean detectors in transition in [920.0, 1000.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 4
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 1000
  Clean detectors in transition in [920.0, 1000.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 5
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 1400
  Clean detectors in transition in [1320.0, 1400.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []
```

```
Column 6
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 740
  Clean detectors in transition in [660.0, 740.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 7
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 400
  Clean detectors in transition in [320.0, 400.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 8
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 450
  Clean detectors in transition in [370.0, 450.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 9
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 325
  Clean detectors in transition in [245.0, 325.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 10
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
```

```
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 1100
  Clean detectors in transition in [1020.0, 1100.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 11
  Noisy detectors excluded: [(0, 11), (1, 11), (3, 11), (5, 11), (6, 11), (8, 11), (9, 11), (10, 11)]
    row=0 col=11 noisy_transition=[895.0, 894.0] min_slope=-1.791 mean_slope=-1.682 drop_kdac=-3.647
    row=0 col=11 noisy_transition=[526.0, 511.0] min_slope=-7.145 mean_slope=-3.258 drop_kdac=-52.710
    row=0 col=11 noisy_transition=[476.0, 475.0] min_slope=-1.940 mean_slope=-1.904 drop_kdac=-3.760
    row=1 col=11 noisy_transition=[895.0, 894.0] min_slope=-1.605 mean_slope=-1.547 drop_kdac=-3.263
    row=1 col=11 noisy_transition=[512.0, 504.0] min_slope=-2.042 mean_slope=-0.703 drop_kdac=-7.563
    row=1 col=11 noisy_transition=[477.0, 476.0] min_slope=-2.257 mean_slope=-1.203 drop_kdac=-0.328
    row=1 col=11 noisy_transition=[472.0, 470.0] min_slope=-2.281 mean_slope=-1.498 drop_kdac=-4.561
    row=1 col=11 noisy_transition=[269.0, 230.0] min_slope=-4.831 mean_slope=-0.574 drop_kdac=-23.062
    row=1 col=11 noisy_transition=[156.0, 155.0] min_slope=-1.300 mean_slope=-1.282 drop_kdac=-2.777
    row=1 col=11 noisy_transition=[133.0, 129.0] min_slope=-9.907 mean_slope=-5.038 drop_kdac=-25.391
    row=1 col=11 noisy_transition=[71.0, 70.0] min_slope=-4.030 mean_slope=-3.008 drop_kdac=-4.536
    row=1 col=11 noisy_transition=[54.0, 52.0] min_slope=-2.858 mean_slope=-1.819 drop_kdac=-5.717
    row=3 col=11 noisy_transition=[1371.0, 1369.0] min_slope=-1.676 mean_slope=-1.083 drop_kdac=-3.352
    row=3 col=11 noisy_transition=[477.0, 476.0] min_slope=-1.627 mean_slope=-0.981 drop_kdac=-0.720
    row=3 col=11 noisy_transition=[472.0, 470.0] min_slope=-1.652 mean_slope=-1.090 drop_kdac=-3.304
    row=5 col=11 noisy_transition=[1370.0, 1369.0] min_slope=-1.627 mean_slope=-1.557 drop_kdac=-3.032
    row=5 col=11 noisy_transition=[472.0, 470.0] min_slope=-1.369 mean_slope=-0.904 drop_kdac=-2.738
    row=6 col=11 noisy_transition=[1313.0, 1308.0] min_slope=-2.152 mean_slope=-1.168 drop_kdac=-7.207
    row=6 col=11 noisy_transition=[896.0, 893.0] min_slope=-3.672 mean_slope=-2.278 drop_kdac=-9.169
    row=6 col=11 noisy_transition=[864.0, 863.0] min_slope=-1.372 mean_slope=-1.267 drop_kdac=-2.806
    row=6 col=11 noisy_transition=[471.0, 470.0] min_slope=-1.741 mean_slope=-1.682 drop_kdac=-3.304
    row=6 col=11 noisy_transition=[206.0, 183.0] min_slope=-5.577 mean_slope=-0.882 drop_kdac=-21.216
    row=6 col=11 noisy_transition=[160.0, 146.0] min_slope=-3.165 mean_slope=-1.976 drop_kdac=-29.922
    row=6 col=11 noisy_transition=[140.0, 139.0] min_slope=-4.651 mean_slope=-4.442 drop_kdac=-9.516
    row=6 col=11 noisy_transition=[133.0, 129.0] min_slope=-1.338 mean_slope=-0.586 drop_kdac=-3.121
    row=6 col=11 noisy_transition=[80.0, 78.0] min_slope=-5.046 mean_slope=-3.107 drop_kdac=-10.092
    row=6 col=11 noisy_transition=[72.0, 70.0] min_slope=-3.015 mean_slope=-1.879 drop_kdac=-6.029
    row=6 col=11 noisy_transition=[53.0, 52.0] min_slope=-1.249 mean_slope=-1.001 drop_kdac=-2.657
    row=8 col=11 noisy_transition=[1439.0, 1434.0] min_slope=-1.928 mean_slope=-0.590 drop_kdac=-3.933
    row=8 col=11 noisy_transition=[1398.0, 1396.0] min_slope=-1.127 mean_slope=-0.818 drop_kdac=-1.564
    row=8 col=11 noisy_transition=[1370.0, 1368.0] min_slope=-3.705 mean_slope=-2.433 drop_kdac=-7.410
    row=8 col=11 noisy_transition=[1314.0, 1308.0] min_slope=-4.366 mean_slope=-1.732 drop_kdac=-12.735
    row=8 col=11 noisy_transition=[898.0, 893.0] min_slope=-3.909 mean_slope=-2.207 drop_kdac=-13.198
    row=8 col=11 noisy_transition=[764.0, 759.0] min_slope=-2.800 mean_slope=-1.186 drop_kdac=-7.343
    row=8 col=11 noisy_transition=[754.0, 751.0] min_slope=-1.758 mean_slope=-0.793 drop_kdac=-2.782
    row=8 col=11 noisy_transition=[522.0, 521.0] min_slope=-1.409 mean_slope=-1.114 drop_kdac=-1.534
    row=8 col=11 noisy_transition=[471.0, 470.0] min_slope=-1.686 mean_slope=-1.662 drop_kdac=-3.341
```

```
    row=9 col=11 noisy_transition=[895.0, 894.0] min_slope=-1.732 mean_slope=-1.627 drop_kdac=-3.464
    row=9 col=11 noisy_transition=[525.0, 521.0] min_slope=-1.454 mean_slope=-0.770 drop_kdac=-3.216
    row=9 col=11 noisy_transition=[516.0, 515.0] min_slope=-1.102 mean_slope=-0.824 drop_kdac=-1.033
    row=9 col=11 noisy_transition=[512.0, 505.0] min_slope=-2.337 mean_slope=-1.207 drop_kdac=-9.787
    row=9 col=11 noisy_transition=[476.0, 476.0] min_slope=-1.178 mean_slope=-1.178 drop_kdac=0.000
    row=9 col=11 noisy_transition=[471.0, 470.0] min_slope=-1.649 mean_slope=-1.645 drop_kdac=-3.259
    row=10 col=11 noisy_transition=[1370.0, 1369.0] min_slope=-4.777 mean_slope=-4.674 drop_kdac=-9.674
    row=10 col=11 noisy_transition=[1311.0, 1308.0] min_slope=-3.166 mean_slope=-1.053 drop_kdac=-3.338
    row=10 col=11 noisy_transition=[896.0, 894.0] min_slope=-1.778 mean_slope=-1.116 drop_kdac=-3.555
    row=10 col=11 noisy_transition=[762.0, 760.0] min_slope=-1.533 mean_slope=-1.006 drop_kdac=-3.066
    row=10 col=11 noisy_transition=[573.0, 562.0] min_slope=-1.115 mean_slope=-0.278 drop_kdac=-3.423
    row=10 col=11 noisy_transition=[532.0, 529.0] min_slope=-5.814 mean_slope=-2.896 drop_kdac=-14.562
    row=10 col=11 noisy_transition=[526.0, 519.0] min_slope=-4.495 mean_slope=-2.350 drop_kdac=-18.994
    row=10 col=11 noisy_transition=[516.0, 511.0] min_slope=-7.610 mean_slope=-4.219 drop_kdac=-25.268
    row=10 col=11 noisy_transition=[472.0, 470.0] min_slope=-2.512 mean_slope=-1.605 drop_kdac=-5.023
    row=10 col=11 noisy_transition=[455.0, 453.0] min_slope=-1.326 mean_slope=-0.804 drop_kdac=-2.652
    row=10 col=11 noisy_transition=[267.0, 263.0] min_slope=-10.503 mean_slope=-5.792 drop_kdac=-30.469
    row=10 col=11 noisy_transition=[160.0, 154.0] min_slope=-3.774 mean_slope=-1.548 drop_kdac=-11.107
    row=10 col=11 noisy_transition=[140.0, 137.0] min_slope=-8.117 mean_slope=-4.636 drop_kdac=-18.756
    row=10 col=11 noisy_transition=[133.0, 132.0] min_slope=-1.500 mean_slope=-1.316 drop_kdac=-3.238
    row=10 col=11 noisy_transition=[81.0, 80.0] min_slope=-1.131 mean_slope=-1.114 drop_kdac=-1.908
    row=10 col=11 noisy_transition=[74.0, 70.0] min_slope=-3.758 mean_slope=-2.469 drop_kdac=-12.825
    row=10 col=11 noisy_transition=[56.0, 52.0] min_slope=-5.573 mean_slope=-3.681 drop_kdac=-18.695
  Best bias = 1798.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [1718.0, 1798.0]: [(13, 11), (14, 11), (15, 11)]
    row=13 col=11 transition=[1783.0, 1770.0] min_slope=-4.313 mean_slope=-2.141 drop_kdac=-29.940
    row=14 col=11 transition=[1778.0, 1771.0] min_slope=-2.731 mean_slope=-1.476 drop_kdac=-11.553
    row=15 col=11 transition=[1729.0, 1718.0] min_slope=-5.000 mean_slope=-2.084 drop_kdac=-24.871
  User bias = 600
  Clean detectors in transition in [520.0, 600.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=0 col=11 transition=[526.0, 511.0] min_slope=-7.145 mean_slope=-3.258 drop_kdac=-52.710
gap_to_window=9.0 reason=excluded as noisy (multiple accepted transitions)
    row=1 col=11 transition=[512.0, 504.0] min_slope=-2.042 mean_slope=-0.703 drop_kdac=-7.563
gap_to_window=16.0 reason=excluded as noisy (multiple accepted transitions)
    row=8 col=11 transition=[522.0, 521.0] min_slope=-1.409 mean_slope=-1.114 drop_kdac=-1.534
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=9 col=11 transition=[525.0, 521.0] min_slope=-1.454 mean_slope=-0.770 drop_kdac=-3.216
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=9 col=11 transition=[516.0, 515.0] min_slope=-1.102 mean_slope=-0.824 drop_kdac=-1.033
gap_to_window=5.0 reason=excluded as noisy (multiple accepted transitions)
    row=9 col=11 transition=[512.0, 505.0] min_slope=-2.337 mean_slope=-1.207 drop_kdac=-9.787
gap_to_window=15.0 reason=excluded as noisy (multiple accepted transitions)
    row=10 col=11 transition=[573.0, 562.0] min_slope=-1.115 mean_slope=-0.278 drop_kdac=-3.423
```

```
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=10 col=11 transition=[532.0, 529.0] min_slope=-5.814 mean_slope=-2.896 drop_kdac=-14.562
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=10 col=11 transition=[526.0, 519.0] min_slope=-4.495 mean_slope=-2.350 drop_kdac=-18.994
gap_to_window=1.0 reason=excluded as noisy (multiple accepted transitions)
    row=10 col=11 transition=[516.0, 511.0] min_slope=-7.610 mean_slope=-4.219 drop_kdac=-25.268
gap_to_window=9.0 reason=excluded as noisy (multiple accepted transitions)

Column 12
  Noisy detectors excluded: []
  Best bias = 174.0
  Max detectors in transition at best bias = 4
  Detectors at best bias in [94.0, 174.0]: [(8, 12), (15, 12), (20, 12), (24, 12)]
    row=8 col=12 transition=[165.0, 140.0] min_slope=-2.636 mean_slope=-0.441 drop_kdac=-11.535
    row=15 col=12 transition=[167.0, 143.0] min_slope=-5.492 mean_slope=-0.685 drop_kdac=-17.325
    row=20 col=12 transition=[173.0, 147.0] min_slope=-4.792 mean_slope=-0.651 drop_kdac=-17.738
    row=24 col=12 transition=[115.0, 94.0] min_slope=-2.491 mean_slope=-0.504 drop_kdac=-11.157
  User bias = 650
  Clean detectors in transition in [570.0, 650.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 13
  Noisy detectors excluded: [(0, 13), (1, 13), (3, 13), (4, 13), (5, 13), (6, 13), (7, 13), (8, 13), (9, 13),
(10, 13), (11, 13), (12, 13), (13, 13), (14, 13)]
    row=0 col=13 noisy_transition=[1329.0, 1324.0] min_slope=-1.266 mean_slope=-0.590 drop_kdac=-3.522
    row=0 col=13 noisy_transition=[838.0, 836.0] min_slope=-2.094 mean_slope=-1.413 drop_kdac=-4.188
    row=0 col=13 noisy_transition=[530.0, 529.0] min_slope=-1.867 mean_slope=-1.783 drop_kdac=-3.641
    row=0 col=13 noisy_transition=[233.0, 229.0] min_slope=-2.119 mean_slope=-0.917 drop_kdac=-4.545
    row=1 col=13 noisy_transition=[993.0, 991.0] min_slope=-2.226 mean_slope=-1.494 drop_kdac=-4.453
    row=1 col=13 noisy_transition=[603.0, 600.0] min_slope=-2.206 mean_slope=-1.135 drop_kdac=-4.499
    row=1 col=13 noisy_transition=[300.0, 294.0] min_slope=-2.152 mean_slope=-0.667 drop_kdac=-4.647
    row=3 col=13 noisy_transition=[961.0, 959.0] min_slope=-2.208 mean_slope=-1.463 drop_kdac=-4.417
    row=3 col=13 noisy_transition=[641.0, 639.0] min_slope=-2.208 mean_slope=-1.465 drop_kdac=-4.415
    row=3 col=13 noisy_transition=[337.0, 335.0] min_slope=-2.254 mean_slope=-1.501 drop_kdac=-4.508
    row=3 col=13 noisy_transition=[47.0, 45.0] min_slope=-2.209 mean_slope=-1.465 drop_kdac=-4.418
    row=4 col=13 noisy_transition=[1289.0, 1287.0] min_slope=-2.186 mean_slope=-1.466 drop_kdac=-4.371
    row=4 col=13 noisy_transition=[855.0, 853.0] min_slope=-2.152 mean_slope=-1.435 drop_kdac=-4.304
    row=4 col=13 noisy_transition=[540.0, 538.0] min_slope=-2.222 mean_slope=-1.483 drop_kdac=-4.444
    row=4 col=13 noisy_transition=[220.0, 218.0] min_slope=-2.213 mean_slope=-1.474 drop_kdac=-4.426
    row=4 col=13 noisy_transition=[144.0, 141.0] min_slope=-1.109 mean_slope=-0.674 drop_kdac=-2.440
    row=4 col=13 noisy_transition=[61.0, 59.0] min_slope=-2.184 mean_slope=-1.445 drop_kdac=-4.368
    row=5 col=13 noisy_transition=[1243.0, 1241.0] min_slope=-2.027 mean_slope=-1.344 drop_kdac=-4.053
    row=5 col=13 noisy_transition=[812.0, 810.0] min_slope=-2.169 mean_slope=-1.429 drop_kdac=-4.339
    row=5 col=13 noisy_transition=[495.0, 493.0] min_slope=-2.075 mean_slope=-1.379 drop_kdac=-4.150
```

row=5 col=13 noisy_transition=[181.0, 179.0] min_slope=-1.894 mean_slope=-1.272 drop_kdac=-3.789
row=6 col=13 noisy_transition=[1188.0, 1186.0] min_slope=-2.086 mean_slope=-1.375 drop_kdac=-4.172
row=6 col=13 noisy_transition=[795.0, 793.0] min_slope=-1.925 mean_slope=-1.275 drop_kdac=-3.849
row=6 col=13 noisy_transition=[488.0, 486.0] min_slope=-1.849 mean_slope=-1.233 drop_kdac=-3.698
row=6 col=13 noisy_transition=[188.0, 186.0] min_slope=-1.910 mean_slope=-1.274 drop_kdac=-3.820
row=7 col=13 noisy_transition=[909.0, 907.0] min_slope=-2.124 mean_slope=-1.411 drop_kdac=-4.249
row=7 col=13 noisy_transition=[590.0, 588.0] min_slope=-2.178 mean_slope=-1.452 drop_kdac=-4.357
row=7 col=13 noisy_transition=[277.0, 275.0] min_slope=-2.142 mean_slope=-1.431 drop_kdac=-4.283
row=7 col=13 noisy_transition=[113.0, 109.0] min_slope=-1.499 mean_slope=-0.554 drop_kdac=-3.104
row=7 col=13 noisy_transition=[36.0, 34.0] min_slope=-2.097 mean_slope=-1.372 drop_kdac=-4.193
row=8 col=13 noisy_transition=[1202.0, 1200.0] min_slope=-2.152 mean_slope=-1.430 drop_kdac=-4.304
row=8 col=13 noisy_transition=[773.0, 771.0] min_slope=-2.210 mean_slope=-1.469 drop_kdac=-4.420
row=8 col=13 noisy_transition=[451.0, 449.0] min_slope=-2.250 mean_slope=-1.488 drop_kdac=-4.501
row=8 col=13 noisy_transition=[64.0, 62.0] min_slope=-2.193 mean_slope=-1.412 drop_kdac=-4.386
row=9 col=13 noisy_transition=[1000.0, 998.0] min_slope=-2.164 mean_slope=-1.440 drop_kdac=-4.329
row=9 col=13 noisy_transition=[669.0, 667.0] min_slope=-2.176 mean_slope=-1.438 drop_kdac=-4.353
row=9 col=13 noisy_transition=[340.0, 338.0] min_slope=-2.215 mean_slope=-1.480 drop_kdac=-4.429
row=9 col=13 noisy_transition=[162.0, 160.0] min_slope=-1.597 mean_slope=-1.066 drop_kdac=-3.193
row=9 col=13 noisy_transition=[78.0, 76.0] min_slope=-2.118 mean_slope=-1.389 drop_kdac=-4.236
row=10 col=13 noisy_transition=[1232.0, 1230.0] min_slope=-2.196 mean_slope=-1.477 drop_kdac=-4.392
row=10 col=13 noisy_transition=[797.0, 796.0] min_slope=-2.100 mean_slope=-1.888 drop_kdac=-3.707
row=10 col=13 noisy_transition=[469.0, 467.0] min_slope=-2.231 mean_slope=-1.514 drop_kdac=-4.462
row=10 col=13 noisy_transition=[147.0, 145.0] min_slope=-2.495 mean_slope=-1.662 drop_kdac=-4.989
row=11 col=13 noisy_transition=[935.0, 932.0] min_slope=-2.130 mean_slope=-1.090 drop_kdac=-4.296
row=11 col=13 noisy_transition=[608.0, 606.0] min_slope=-2.156 mean_slope=-1.446 drop_kdac=-4.311
row=11 col=13 noisy_transition=[279.0, 278.0] min_slope=-2.175 mean_slope=-2.051 drop_kdac=-3.837
row=11 col=13 noisy_transition=[178.0, 177.0] min_slope=-1.450 mean_slope=-1.227 drop_kdac=-1.640
row=11 col=13 noisy_transition=[93.0, 91.0] min_slope=-2.011 mean_slope=-1.290 drop_kdac=-4.023
row=11 col=13 noisy_transition=[13.0, 11.0] min_slope=-2.068 mean_slope=-1.375 drop_kdac=-4.135
row=12 col=13 noisy_transition=[1731.0, 1729.0] min_slope=-2.119 mean_slope=-1.384 drop_kdac=-4.238
row=12 col=13 noisy_transition=[1657.0, 1655.0] min_slope=-2.085 mean_slope=-1.374 drop_kdac=-4.170
row=12 col=13 noisy_transition=[1582.0, 1580.0] min_slope=-2.122 mean_slope=-1.409 drop_kdac=-4.245
row=12 col=13 noisy_transition=[1507.0, 1505.0] min_slope=-2.114 mean_slope=-1.392 drop_kdac=-4.227
row=12 col=13 noisy_transition=[1432.0, 1430.0] min_slope=-1.979 mean_slope=-1.294 drop_kdac=-3.958
row=12 col=13 noisy_transition=[1357.0, 1355.0] min_slope=-1.987 mean_slope=-1.304 drop_kdac=-3.975
row=12 col=13 noisy_transition=[1286.0, 1280.0] min_slope=-2.002 mean_slope=-0.696 drop_kdac=-4.815
row=12 col=13 noisy_transition=[1208.0, 1206.0] min_slope=-1.999 mean_slope=-1.315 drop_kdac=-3.998
row=12 col=13 noisy_transition=[1133.0, 1131.0] min_slope=-2.024 mean_slope=-1.331 drop_kdac=-4.048
row=12 col=13 noisy_transition=[1058.0, 1056.0] min_slope=-2.088 mean_slope=-1.371 drop_kdac=-4.176
row=12 col=13 noisy_transition=[984.0, 982.0] min_slope=-2.130 mean_slope=-1.404 drop_kdac=-4.261
row=12 col=13 noisy_transition=[909.0, 906.0] min_slope=-1.573 mean_slope=-0.882 drop_kdac=-3.500
row=12 col=13 noisy_transition=[834.0, 832.0] min_slope=-2.098 mean_slope=-1.358 drop_kdac=-4.195
row=12 col=13 noisy_transition=[760.0, 758.0] min_slope=-2.108 mean_slope=-1.390 drop_kdac=-4.216
row=12 col=13 noisy_transition=[685.0, 683.0] min_slope=-2.144 mean_slope=-1.413 drop_kdac=-4.287
row=12 col=13 noisy_transition=[610.0, 606.0] min_slope=-2.122 mean_slope=-0.892 drop_kdac=-4.515

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    row=12 col=13 noisy_transition=[537.0, 534.0] min_slope=-2.355 mean_slope=-1.312 drop_kdac=-5.181
    row=12 col=13 noisy_transition=[461.0, 459.0] min_slope=-2.094 mean_slope=-1.388 drop_kdac=-4.187
    row=12 col=13 noisy_transition=[386.0, 384.0] min_slope=-2.071 mean_slope=-1.380 drop_kdac=-4.142
    row=12 col=13 noisy_transition=[312.0, 310.0] min_slope=-2.103 mean_slope=-1.382 drop_kdac=-4.206
    row=12 col=13 noisy_transition=[237.0, 235.0] min_slope=-2.025 mean_slope=-1.320 drop_kdac=-4.049
    row=12 col=13 noisy_transition=[162.0, 159.0] min_slope=-1.844 mean_slope=-0.964 drop_kdac=-4.053
    row=12 col=13 noisy_transition=[146.0, 144.0] min_slope=-1.010 mean_slope=-0.437 drop_kdac=-2.019
    row=12 col=13 noisy_transition=[87.0, 85.0] min_slope=-1.920 mean_slope=-1.220 drop_kdac=-3.840
    row=12 col=13 noisy_transition=[13.0, 11.0] min_slope=-2.329 mean_slope=-1.564 drop_kdac=-4.658
    row=13 col=13 noisy_transition=[899.0, 894.0] min_slope=-2.219 mean_slope=-0.797 drop_kdac=-4.708
    row=13 col=13 noisy_transition=[570.0, 568.0] min_slope=-2.331 mean_slope=-1.555 drop_kdac=-4.662
    row=13 col=13 noisy_transition=[242.0, 240.0] min_slope=-2.229 mean_slope=-1.481 drop_kdac=-4.457
    row=13 col=13 noisy_transition=[104.0, 102.0] min_slope=-1.561 mean_slope=-1.006 drop_kdac=-3.122
    row=13 col=13 noisy_transition=[25.0, 23.0] min_slope=-2.261 mean_slope=-1.493 drop_kdac=-4.523
    row=14 col=13 noisy_transition=[809.0, 805.0] min_slope=-2.393 mean_slope=-1.184 drop_kdac=-5.906
    row=14 col=13 noisy_transition=[474.0, 472.0] min_slope=-2.385 mean_slope=-1.593 drop_kdac=-4.771
    row=14 col=13 noisy_transition=[151.0, 148.0] min_slope=-2.385 mean_slope=-1.230 drop_kdac=-4.805
    row=14 col=13 noisy_transition=[11.0, 10.0] min_slope=-1.025 mean_slope=-0.663 drop_kdac=-0.559
  Best bias = 223.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [143.0, 223.0]: [(15, 13)]
    row=15 col=13 transition=[149.0, 143.0] min_slope=-1.573 mean_slope=-0.605 drop_kdac=-4.576
  User bias = 1000
  Clean detectors in transition in [920.0, 1000.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=1 col=13 transition=[993.0, 991.0] min_slope=-2.226 mean_slope=-1.494 drop_kdac=-4.453
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=3 col=13 transition=[961.0, 959.0] min_slope=-2.208 mean_slope=-1.463 drop_kdac=-4.417
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=7 col=13 transition=[909.0, 907.0] min_slope=-2.124 mean_slope=-1.411 drop_kdac=-4.249
gap_to_window=13.0 reason=excluded as noisy (multiple accepted transitions)
    row=9 col=13 transition=[1000.0, 998.0] min_slope=-2.164 mean_slope=-1.440 drop_kdac=-4.329
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=11 col=13 transition=[935.0, 932.0] min_slope=-2.130 mean_slope=-1.090 drop_kdac=-4.296
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=12 col=13 transition=[984.0, 982.0] min_slope=-2.130 mean_slope=-1.404 drop_kdac=-4.261
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=12 col=13 transition=[909.0, 906.0] min_slope=-1.573 mean_slope=-0.882 drop_kdac=-3.500
gap_to_window=14.0 reason=excluded as noisy (multiple accepted transitions)
    row=13 col=13 transition=[899.0, 894.0] min_slope=-2.219 mean_slope=-0.797 drop_kdac=-4.708
gap_to_window=26.0 reason=excluded as noisy (multiple accepted transitions)
    row=18 col=13 transition=[943.0, 943.0] min_slope=-0.101 mean_slope=-0.101 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=19 col=13 transition=[928.0, 927.0] min_slope=-0.145 mean_slope=-0.124 drop_kdac=-0.289
```

gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=21 col=13 transition=[1037.0, 1036.0] min_slope=-0.142 mean_slope=-0.139 drop_kdac=-0.251
gap_to_window=37.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=13 transition=[1035.0, 1035.0] min_slope=-0.090 mean_slope=-0.090 drop_kdac=0.000
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=13 transition=[1029.0, 1029.0] min_slope=-0.085 mean_slope=-0.085 drop_kdac=0.000
gap_to_window=29.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=13 transition=[1026.0, 1026.0] min_slope=-0.104 mean_slope=-0.104 drop_kdac=0.000
gap_to_window=26.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=13 transition=[1012.0, 1012.0] min_slope=-0.094 mean_slope=-0.094 drop_kdac=0.000
gap_to_window=12.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=13 transition=[1009.0, 1009.0] min_slope=-0.089 mean_slope=-0.089 drop_kdac=0.000
gap_to_window=9.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=13 transition=[1006.0, 1006.0] min_slope=-0.099 mean_slope=-0.099 drop_kdac=0.000
gap_to_window=6.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=13 transition=[953.0, 953.0] min_slope=-0.095 mean_slope=-0.095 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=13 transition=[936.0, 936.0] min_slope=-0.086 mean_slope=-0.086 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=26 col=13 transition=[989.0, 989.0] min_slope=-0.080 mean_slope=-0.080 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=26 col=13 transition=[984.0, 984.0] min_slope=-0.095 mean_slope=-0.095 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=26 col=13 transition=[970.0, 969.0] min_slope=-0.120 mean_slope=-0.103 drop_kdac=-0.117
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=26 col=13 transition=[966.0, 966.0] min_slope=-0.107 mean_slope=-0.107 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=26 col=13 transition=[957.0, 957.0] min_slope=-0.107 mean_slope=-0.107 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=26 col=13 transition=[935.0, 935.0] min_slope=-0.089 mean_slope=-0.089 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=26 col=13 transition=[889.0, 888.0] min_slope=-0.190 mean_slope=-0.150 drop_kdac=-0.225

gap_to_window=32.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=27 col=13 transition=[1037.0, 1037.0] min_slope=-0.096 mean_slope=-0.096 drop_kdac=0.000
gap_to_window=37.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[1034.0, 1034.0] min_slope=-0.137 mean_slope=-0.137 drop_kdac=0.000
gap_to_window=34.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[1031.0, 1031.0] min_slope=-0.114 mean_slope=-0.114 drop_kdac=0.000
gap_to_window=31.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[1028.0, 1028.0] min_slope=-0.124 mean_slope=-0.124 drop_kdac=0.000
gap_to_window=28.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[1024.0, 1023.0] min_slope=-0.100 mean_slope=-0.098 drop_kdac=-0.147
gap_to_window=24.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=27 col=13 transition=[1020.0, 1020.0] min_slope=-0.080 mean_slope=-0.080 drop_kdac=0.000
gap_to_window=20.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[1017.0, 1017.0] min_slope=-0.110 mean_slope=-0.110 drop_kdac=0.000
gap_to_window=17.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[999.0, 999.0] min_slope=-0.140 mean_slope=-0.140 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[983.0, 983.0] min_slope=-0.092 mean_slope=-0.092 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[976.0, 976.0] min_slope=-0.096 mean_slope=-0.096 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[972.0, 972.0] min_slope=-0.081 mean_slope=-0.081 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[958.0, 958.0] min_slope=-0.096 mean_slope=-0.096 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[939.0, 939.0] min_slope=-0.094 mean_slope=-0.094 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[899.0, 899.0] min_slope=-0.103 mean_slope=-0.103 drop_kdac=0.000
gap_to_window=21.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=27 col=13 transition=[895.0, 895.0] min_slope=-0.081 mean_slope=-0.081 drop_kdac=0.000
gap_to_window=25.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

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    row=27 col=13 transition=[888.0, 888.0] min_slope=-0.166 mean_slope=-0.166 drop_kdac=0.000
gap_to_window=32.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[1033.0, 1033.0] min_slope=-0.133 mean_slope=-0.133 drop_kdac=0.000
gap_to_window=33.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[1028.0, 1028.0] min_slope=-0.233 mean_slope=-0.233 drop_kdac=0.000
gap_to_window=28.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[1021.0, 1021.0] min_slope=-0.156 mean_slope=-0.156 drop_kdac=0.000
gap_to_window=21.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[1018.0, 1018.0] min_slope=-0.091 mean_slope=-0.091 drop_kdac=0.000
gap_to_window=18.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[1003.0, 1003.0] min_slope=-0.088 mean_slope=-0.088 drop_kdac=0.000
gap_to_window=3.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[998.0, 998.0] min_slope=-0.103 mean_slope=-0.103 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[990.0, 990.0] min_slope=-0.121 mean_slope=-0.121 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[982.0, 982.0] min_slope=-0.131 mean_slope=-0.131 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[965.0, 965.0] min_slope=-0.175 mean_slope=-0.175 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[951.0, 951.0] min_slope=-0.121 mean_slope=-0.121 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[943.0, 943.0] min_slope=-0.117 mean_slope=-0.117 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[939.0, 939.0] min_slope=-0.080 mean_slope=-0.080 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[935.0, 935.0] min_slope=-0.163 mean_slope=-0.163 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[931.0, 931.0] min_slope=-0.093 mean_slope=-0.093 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
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    row=28 col=13 transition=[916.0, 916.0] min_slope=-0.122 mean_slope=-0.122 drop_kdac=0.000
gap_to_window=4.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[912.0, 912.0] min_slope=-0.100 mean_slope=-0.100 drop_kdac=0.000
gap_to_window=8.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[906.0, 906.0] min_slope=-0.082 mean_slope=-0.082 drop_kdac=0.000
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[903.0, 903.0] min_slope=-0.081 mean_slope=-0.081 drop_kdac=0.000
gap_to_window=17.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[899.0, 899.0] min_slope=-0.118 mean_slope=-0.118 drop_kdac=0.000
gap_to_window=21.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[893.0, 893.0] min_slope=-0.248 mean_slope=-0.248 drop_kdac=0.000
gap_to_window=27.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=28 col=13 transition=[888.0, 888.0] min_slope=-0.121 mean_slope=-0.121 drop_kdac=0.000
gap_to_window=32.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[1040.0, 1040.0] min_slope=-0.094 mean_slope=-0.094 drop_kdac=0.000
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[1036.0, 1036.0] min_slope=-0.123 mean_slope=-0.123 drop_kdac=0.000
gap_to_window=36.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[1033.0, 1032.0] min_slope=-0.166 mean_slope=-0.124 drop_kdac=-0.483
gap_to_window=33.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=13 transition=[1028.0, 1027.0] min_slope=-0.131 mean_slope=-0.118 drop_kdac=-0.324
gap_to_window=28.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=13 transition=[1024.0, 1024.0] min_slope=-0.118 mean_slope=-0.118 drop_kdac=0.000
gap_to_window=24.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[1021.0, 1021.0] min_slope=-0.161 mean_slope=-0.161 drop_kdac=0.000
gap_to_window=21.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[1015.0, 1014.0] min_slope=-0.130 mean_slope=-0.121 drop_kdac=-0.211
gap_to_window=15.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=13 transition=[1011.0, 1011.0] min_slope=-0.204 mean_slope=-0.204 drop_kdac=0.000
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[1000.0, 999.0] min_slope=-0.122 mean_slope=-0.105 drop_kdac=-0.167
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=13 transition=[982.0, 982.0] min_slope=-0.213 mean_slope=-0.213 drop_kdac=0.000
```

gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[977.0, 977.0] min_slope=-0.120 mean_slope=-0.120 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[959.0, 958.0] min_slope=-0.114 mean_slope=-0.101 drop_kdac=-0.030
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=13 transition=[940.0, 940.0] min_slope=-0.093 mean_slope=-0.093 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[921.0, 921.0] min_slope=-0.130 mean_slope=-0.130 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[912.0, 912.0] min_slope=-0.115 mean_slope=-0.115 drop_kdac=0.000
gap_to_window=8.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[909.0, 909.0] min_slope=-0.123 mean_slope=-0.123 drop_kdac=0.000
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[903.0, 903.0] min_slope=-0.091 mean_slope=-0.091 drop_kdac=0.000
gap_to_window=17.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[892.0, 892.0] min_slope=-0.092 mean_slope=-0.092 drop_kdac=0.000
gap_to_window=28.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[887.0, 887.0] min_slope=-0.126 mean_slope=-0.126 drop_kdac=0.000
gap_to_window=33.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=29 col=13 transition=[881.0, 881.0] min_slope=-0.125 mean_slope=-0.125 drop_kdac=0.000
gap_to_window=39.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[1027.0, 1027.0] min_slope=-0.101 mean_slope=-0.101 drop_kdac=0.000
gap_to_window=27.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[1015.0, 1014.0] min_slope=-0.101 mean_slope=-0.097 drop_kdac=-0.158
gap_to_window=15.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=30 col=13 transition=[1011.0, 1011.0] min_slope=-0.116 mean_slope=-0.116 drop_kdac=0.000
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[1008.0, 1008.0] min_slope=-0.115 mean_slope=-0.115 drop_kdac=0.000
gap_to_window=8.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[998.0, 998.0] min_slope=-0.128 mean_slope=-0.128 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

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    row=30 col=13 transition=[990.0, 990.0] min_slope=-0.087 mean_slope=-0.087 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[976.0, 976.0] min_slope=-0.095 mean_slope=-0.095 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[939.0, 939.0] min_slope=-0.086 mean_slope=-0.086 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[934.0, 934.0] min_slope=-0.114 mean_slope=-0.114 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[927.0, 927.0] min_slope=-0.104 mean_slope=-0.104 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[921.0, 921.0] min_slope=-0.090 mean_slope=-0.090 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=30 col=13 transition=[887.0, 887.0] min_slope=-0.108 mean_slope=-0.108 drop_kdac=0.000
gap_to_window=33.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[1040.0, 1039.0] min_slope=-0.165 mean_slope=-0.148 drop_kdac=-0.273
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=31 col=13 transition=[1035.0, 1035.0] min_slope=-0.087 mean_slope=-0.087 drop_kdac=0.000
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[1027.0, 1027.0] min_slope=-0.142 mean_slope=-0.142 drop_kdac=0.000
gap_to_window=27.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[1023.0, 1023.0] min_slope=-0.084 mean_slope=-0.084 drop_kdac=0.000
gap_to_window=23.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[1014.0, 1014.0] min_slope=-0.098 mean_slope=-0.098 drop_kdac=0.000
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[998.0, 997.0] min_slope=-0.111 mean_slope=-0.107 drop_kdac=-0.173
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=31 col=13 transition=[994.0, 994.0] min_slope=-0.118 mean_slope=-0.118 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[989.0, 989.0] min_slope=-0.091 mean_slope=-0.091 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[977.0, 976.0] min_slope=-0.098 mean_slope=-0.092 drop_kdac=-0.133
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
```

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    row=31 col=13 transition=[946.0, 946.0] min_slope=-0.087 mean_slope=-0.087 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[935.0, 934.0] min_slope=-0.103 mean_slope=-0.094 drop_kdac=-0.071
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=31 col=13 transition=[927.0, 927.0] min_slope=-0.100 mean_slope=-0.100 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[920.0, 920.0] min_slope=-0.178 mean_slope=-0.178 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[911.0, 911.0] min_slope=-0.099 mean_slope=-0.099 drop_kdac=0.000
gap_to_window=9.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=31 col=13 transition=[887.0, 886.0] min_slope=-0.121 mean_slope=-0.115 drop_kdac=-0.131
gap_to_window=34.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=13 transition=[1027.0, 1027.0] min_slope=-0.148 mean_slope=-0.148 drop_kdac=0.000
gap_to_window=27.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[1023.0, 1023.0] min_slope=-0.105 mean_slope=-0.105 drop_kdac=0.000
gap_to_window=23.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[1020.0, 1020.0] min_slope=-0.083 mean_slope=-0.083 drop_kdac=0.000
gap_to_window=20.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[1014.0, 1014.0] min_slope=-0.081 mean_slope=-0.081 drop_kdac=0.000
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[1002.0, 1002.0] min_slope=-0.105 mean_slope=-0.105 drop_kdac=0.000
gap_to_window=2.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[998.0, 997.0] min_slope=-0.157 mean_slope=-0.123 drop_kdac=-0.087
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=13 transition=[994.0, 994.0] min_slope=-0.095 mean_slope=-0.095 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[988.0, 988.0] min_slope=-0.160 mean_slope=-0.160 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[976.0, 976.0] min_slope=-0.113 mean_slope=-0.113 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[972.0, 972.0] min_slope=-0.113 mean_slope=-0.113 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
```

```
    row=32 col=13 transition=[964.0, 964.0] min_slope=-0.101 mean_slope=-0.101 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[957.0, 957.0] min_slope=-0.124 mean_slope=-0.124 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[939.0, 937.0] min_slope=-0.100 mean_slope=-0.091 drop_kdac=-0.200
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=13 transition=[934.0, 934.0] min_slope=-0.167 mean_slope=-0.167 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[923.0, 923.0] min_slope=-0.081 mean_slope=-0.081 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[919.0, 919.0] min_slope=-0.094 mean_slope=-0.094 drop_kdac=0.000
gap_to_window=1.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[914.0, 914.0] min_slope=-0.087 mean_slope=-0.087 drop_kdac=0.000
gap_to_window=6.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[904.0, 904.0] min_slope=-0.104 mean_slope=-0.104 drop_kdac=0.000
gap_to_window=16.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[894.0, 894.0] min_slope=-0.082 mean_slope=-0.082 drop_kdac=0.000
gap_to_window=26.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[890.0, 890.0] min_slope=-0.111 mean_slope=-0.111 drop_kdac=0.000
gap_to_window=30.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=13 transition=[887.0, 886.0] min_slope=-0.115 mean_slope=-0.099 drop_kdac=-0.127
gap_to_window=34.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=13 transition=[881.0, 881.0] min_slope=-0.100 mean_slope=-0.100 drop_kdac=0.000
gap_to_window=39.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

Column 14
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 675
  Clean detectors in transition in [595.0, 675.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []
```

```
Column 15
  Noisy detectors excluded: [(31, 15)]
    row=31 col=15 noisy_transition=[1215.0, 1212.0] min_slope=-2.751 mean_slope=-1.622 drop_kdac=-5.796
    row=31 col=15 noisy_transition=[1159.0, 1157.0] min_slope=-4.039 mean_slope=-2.583 drop_kdac=-8.079
    row=31 col=15 noisy_transition=[1152.0, 1135.0] min_slope=-2.957 mean_slope=-1.215 drop_kdac=-21.816
    row=31 col=15 noisy_transition=[1123.0, 1120.0] min_slope=-1.116 mean_slope=-0.610 drop_kdac=-2.600
    row=31 col=15 noisy_transition=[1095.0, 1094.0] min_slope=-2.403 mean_slope=-2.397 drop_kdac=-4.761
    row=31 col=15 noisy_transition=[1076.0, 1065.0] min_slope=-1.869 mean_slope=-0.897 drop_kdac=-10.688
    row=31 col=15 noisy_transition=[1060.0, 1054.0] min_slope=-1.396 mean_slope=-0.390 drop_kdac=-3.135
    row=31 col=15 noisy_transition=[1048.0, 1038.0] min_slope=-1.709 mean_slope=-0.758 drop_kdac=-8.225
  Best bias = 636.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [556.0, 636.0]: [(1, 15), (6, 15), (7, 15)]
    row=1 col=15 transition=[613.0, 558.0] min_slope=-5.718 mean_slope=-0.578 drop_kdac=-32.418
    row=6 col=15 transition=[579.0, 556.0] min_slope=-8.594 mean_slope=-1.809 drop_kdac=-43.409
    row=7 col=15 transition=[612.0, 559.0] min_slope=-3.401 mean_slope=-0.292 drop_kdac=-15.608
  User bias = 650
  Clean detectors in transition in [570.0, 650.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=1 col=15 transition=[613.0, 558.0] min_slope=-5.718 mean_slope=-0.578 drop_kdac=-32.418
gap_to_window=12.0 reason=accepted transition outside user window
    row=6 col=15 transition=[579.0, 556.0] min_slope=-8.594 mean_slope=-1.809 drop_kdac=-43.409
gap_to_window=14.0 reason=accepted transition outside user window
    row=7 col=15 transition=[612.0, 559.0] min_slope=-3.401 mean_slope=-0.292 drop_kdac=-15.608
gap_to_window=11.0 reason=accepted transition outside user window
    row=8 col=15 transition=[595.0, 562.0] min_slope=-0.886 mean_slope=-0.179 drop_kdac=-6.062
gap_to_window=8.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=15 transition=[621.0, 620.0] min_slope=-0.083 mean_slope=-0.082 drop_kdac=-0.108
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=15 transition=[595.0, 553.0] min_slope=-0.721 mean_slope=-0.193 drop_kdac=-8.223
gap_to_window=17.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 16
  Noisy detectors excluded: [(0, 16), (6, 16), (9, 16), (12, 16), (20, 16), (30, 16)]
    row=0 col=16 noisy_transition=[1342.0, 1341.0] min_slope=-1.435 mean_slope=-1.418 drop_kdac=-2.776
    row=0 col=16 noisy_transition=[861.0, 860.0] min_slope=-1.035 mean_slope=-0.971 drop_kdac=-2.065
    row=6 col=16 noisy_transition=[325.0, 312.0] min_slope=-8.228 mean_slope=-4.105 drop_kdac=-54.646
    row=6 col=16 noisy_transition=[309.0, 307.0] min_slope=-9.238 mean_slope=-4.916 drop_kdac=-10.772
    row=9 col=16 noisy_transition=[1223.0, 1218.0] min_slope=-1.908 mean_slope=-1.054 drop_kdac=-6.219
    row=9 col=16 noisy_transition=[1159.0, 1158.0] min_slope=-1.274 mean_slope=-1.264 drop_kdac=-2.736
    row=9 col=16 noisy_transition=[1150.0, 1142.0] min_slope=-1.437 mean_slope=-0.574 drop_kdac=-5.095
    row=9 col=16 noisy_transition=[1139.0, 1138.0] min_slope=-1.286 mean_slope=-1.253 drop_kdac=-2.417
    row=9 col=16 noisy_transition=[1135.0, 1132.0] min_slope=-1.417 mean_slope=-1.122 drop_kdac=-4.375
    row=9 col=16 noisy_transition=[1121.0, 1114.0] min_slope=-1.561 mean_slope=-0.647 drop_kdac=-5.289
    row=12 col=16 noisy_transition=[544.0, 504.0] min_slope=-2.224 mean_slope=-0.392 drop_kdac=-15.976
```

```
    row=12 col=16 noisy_transition=[492.0, 491.0] min_slope=-1.299 mean_slope=-1.237 drop_kdac=-2.520
    row=20 col=16 noisy_transition=[995.0, 994.0] min_slope=-1.031 mean_slope=-1.013 drop_kdac=-2.061
    row=20 col=16 noisy_transition=[349.0, 348.0] min_slope=-1.077 mean_slope=-1.025 drop_kdac=-2.152
    row=20 col=16 noisy_transition=[101.0, 89.0] min_slope=-1.247 mean_slope=-0.337 drop_kdac=-4.365
    row=20 col=16 noisy_transition=[18.0, 17.0] min_slope=-1.074 mean_slope=-1.007 drop_kdac=-2.139
    row=30 col=16 noisy_transition=[1177.0, 1174.0] min_slope=-5.994 mean_slope=-3.298 drop_kdac=-13.243
    row=30 col=16 noisy_transition=[1170.0, 1169.0] min_slope=-1.191 mean_slope=-1.106 drop_kdac=-2.325
  Best bias = 536.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [456.0, 536.0]: [(3, 16)]
    row=3 col=16 transition=[491.0, 456.0] min_slope=-3.807 mean_slope=-0.343 drop_kdac=-12.405
  User bias = 1450
  Clean detectors in transition in [1370.0, 1450.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=0 col=16 transition=[1342.0, 1341.0] min_slope=-1.435 mean_slope=-1.418 drop_kdac=-2.776
gap_to_window=29.0 reason=excluded as noisy (multiple accepted transitions)
    row=18 col=16 transition=[1344.0, 1343.0] min_slope=-0.824 mean_slope=-0.806 drop_kdac=-1.572
gap_to_window=27.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=22 col=16 transition=[1352.0, 1350.0] min_slope=-0.875 mean_slope=-0.582 drop_kdac=-1.750
gap_to_window=20.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 17
  Noisy detectors excluded: [(4, 17), (12, 17), (14, 17), (21, 17), (22, 17), (23, 17), (29, 17)]
    row=4 col=17 noisy_transition=[988.0, 987.0] min_slope=-1.484 mean_slope=-1.460 drop_kdac=-2.862
    row=4 col=17 noisy_transition=[977.0, 975.0] min_slope=-2.179 mean_slope=-1.453 drop_kdac=-4.358
    row=4 col=17 noisy_transition=[555.0, 554.0] min_slope=-1.302 mean_slope=-1.280 drop_kdac=-2.619
    row=4 col=17 noisy_transition=[490.0, 489.0] min_slope=-1.308 mean_slope=-1.259 drop_kdac=-2.534
    row=4 col=17 noisy_transition=[362.0, 360.0] min_slope=-3.020 mean_slope=-2.011 drop_kdac=-6.041
    row=12 col=17 noisy_transition=[1044.0, 1042.0] min_slope=-1.281 mean_slope=-0.834 drop_kdac=-2.563
    row=12 col=17 noisy_transition=[801.0, 799.0] min_slope=-2.486 mean_slope=-1.683 drop_kdac=-4.971
    row=14 col=17 noisy_transition=[1512.0, 1510.0] min_slope=-4.440 mean_slope=-2.941 drop_kdac=-8.879
    row=14 col=17 noisy_transition=[206.0, 193.0] min_slope=-1.797 mean_slope=-0.390 drop_kdac=-5.465
    row=21 col=17 noisy_transition=[574.0, 573.0] min_slope=-1.350 mean_slope=-1.309 drop_kdac=-2.718
    row=21 col=17 noisy_transition=[134.0, 130.0] min_slope=-1.139 mean_slope=-0.511 drop_kdac=-2.572
    row=22 col=17 noisy_transition=[1773.0, 1769.0] min_slope=-2.574 mean_slope=-1.521 drop_kdac=-7.535
    row=22 col=17 noisy_transition=[1302.0, 1300.0] min_slope=-3.919 mean_slope=-2.631 drop_kdac=-7.838
    row=22 col=17 noisy_transition=[1296.0, 1293.0] min_slope=-3.979 mean_slope=-2.174 drop_kdac=-8.756
    row=22 col=17 noisy_transition=[1287.0, 1285.0] min_slope=-1.230 mean_slope=-0.826 drop_kdac=-2.460
    row=22 col=17 noisy_transition=[1229.0, 1227.0] min_slope=-1.236 mean_slope=-0.820 drop_kdac=-2.472
    row=22 col=17 noisy_transition=[339.0, 338.0] min_slope=-1.292 mean_slope=-1.267 drop_kdac=-2.586
    row=22 col=17 noisy_transition=[315.0, 309.0] min_slope=-3.130 mean_slope=-1.248 drop_kdac=-8.769
    row=22 col=17 noisy_transition=[199.0, 198.0] min_slope=-1.330 mean_slope=-1.245 drop_kdac=-2.583
    row=22 col=17 noisy_transition=[187.0, 181.0] min_slope=-1.348 mean_slope=-0.702 drop_kdac=-4.923
    row=22 col=17 noisy_transition=[177.0, 172.0] min_slope=-3.084 mean_slope=-1.242 drop_kdac=-7.455
    row=22 col=17 noisy_transition=[27.0, 25.0] min_slope=-1.305 mean_slope=-0.836 drop_kdac=-2.610
```

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    row=23 col=17 noisy_transition=[1772.0, 1766.0] min_slope=-1.219 mean_slope=-0.379 drop_kdac=-2.452
    row=23 col=17 noisy_transition=[1746.0, 1745.0] min_slope=-1.455 mean_slope=-1.347 drop_kdac=-2.853
    row=23 col=17 noisy_transition=[342.0, 335.0] min_slope=-2.583 mean_slope=-1.251 drop_kdac=-10.002
    row=23 col=17 noisy_transition=[315.0, 311.0] min_slope=-4.861 mean_slope=-2.770 drop_kdac=-13.954
    row=23 col=17 noisy_transition=[116.0, 115.0] min_slope=-1.263 mean_slope=-1.236 drop_kdac=-2.457
    row=23 col=17 noisy_transition=[85.0, 84.0] min_slope=-1.260 mean_slope=-1.217 drop_kdac=-2.355
    row=23 col=17 noisy_transition=[81.0, 80.0] min_slope=-1.266 mean_slope=-1.261 drop_kdac=-2.568
    row=29 col=17 noisy_transition=[1762.0, 1761.0] min_slope=-1.252 mean_slope=-1.234 drop_kdac=-2.455
    row=29 col=17 noisy_transition=[1435.0, 1433.0] min_slope=-1.255 mean_slope=-0.829 drop_kdac=-2.510
    row=29 col=17 noisy_transition=[1428.0, 1427.0] min_slope=-1.264 mean_slope=-1.259 drop_kdac=-2.547
    row=29 col=17 noisy_transition=[1138.0, 1137.0] min_slope=-1.362 mean_slope=-1.304 drop_kdac=-2.745
    row=29 col=17 noisy_transition=[1101.0, 1100.0] min_slope=-1.353 mean_slope=-1.297 drop_kdac=-2.700
    row=29 col=17 noisy_transition=[922.0, 920.0] min_slope=-1.227 mean_slope=-0.819 drop_kdac=-2.453
    row=29 col=17 noisy_transition=[751.0, 749.0] min_slope=-1.253 mean_slope=-0.826 drop_kdac=-2.506
    row=29 col=17 noisy_transition=[597.0, 596.0] min_slope=-1.252 mean_slope=-1.246 drop_kdac=-2.496
    row=29 col=17 noisy_transition=[470.0, 469.0] min_slope=-1.316 mean_slope=-1.289 drop_kdac=-2.653
    row=29 col=17 noisy_transition=[466.0, 464.0] min_slope=-1.262 mean_slope=-0.832 drop_kdac=-2.525
    row=29 col=17 noisy_transition=[416.0, 415.0] min_slope=-1.239 mean_slope=-1.238 drop_kdac=-2.477
    row=29 col=17 noisy_transition=[88.0, 87.0] min_slope=-1.243 mean_slope=-1.230 drop_kdac=-2.440
  Best bias = 1143.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [1063.0, 1143.0]: [(32, 17)]
    row=32 col=17 transition=[1065.0, 1063.0] min_slope=-1.222 mean_slope=-0.826 drop_kdac=-2.443
  User bias = 800
  Clean detectors in transition in [720.0, 800.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=12 col=17 transition=[801.0, 799.0] min_slope=-2.486 mean_slope=-1.683 drop_kdac=-4.971
gap_to_window=1.0 reason=excluded as noisy (multiple accepted transitions)
    row=29 col=17 transition=[751.0, 749.0] min_slope=-1.253 mean_slope=-0.826 drop_kdac=-2.506
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)

Column 18
  Noisy detectors excluded: [(13, 18), (14, 18), (18, 18), (19, 18), (30, 18)]
    row=13 col=18 noisy_transition=[1487.0, 1486.0] min_slope=-1.020 mean_slope=-0.999 drop_kdac=-2.037
    row=13 col=18 noisy_transition=[1355.0, 1354.0] min_slope=-1.019 mean_slope=-1.007 drop_kdac=-1.986
    row=13 col=18 noisy_transition=[1221.0, 1220.0] min_slope=-1.007 mean_slope=-0.998 drop_kdac=-1.981
    row=13 col=18 noisy_transition=[368.0, 344.0] min_slope=-1.829 mean_slope=-0.283 drop_kdac=-7.158
    row=14 col=18 noisy_transition=[1167.0, 1166.0] min_slope=-1.048 mean_slope=-1.014 drop_kdac=-2.094
    row=14 col=18 noisy_transition=[1027.0, 1026.0] min_slope=-1.026 mean_slope=-0.988 drop_kdac=-2.051
    row=14 col=18 noisy_transition=[607.0, 606.0] min_slope=-1.020 mean_slope=-0.995 drop_kdac=-2.035
    row=14 col=18 noisy_transition=[315.0, 295.0] min_slope=-1.557 mean_slope=-0.296 drop_kdac=-6.199
    row=18 col=18 noisy_transition=[1699.0, 1698.0] min_slope=-1.019 mean_slope=-1.010 drop_kdac=-1.994
    row=18 col=18 noisy_transition=[1532.0, 1531.0] min_slope=-1.062 mean_slope=-1.045 drop_kdac=-2.123
    row=18 col=18 noisy_transition=[1376.0, 1375.0] min_slope=-1.026 mean_slope=-1.016 drop_kdac=-2.007
    row=18 col=18 noisy_transition=[1230.0, 1229.0] min_slope=-1.037 mean_slope=-1.036 drop_kdac=-2.068
```

```
    row=18 col=18 noisy_transition=[1086.0, 1085.0] min_slope=-1.011 mean_slope=-1.001 drop_kdac=-1.980
    row=18 col=18 noisy_transition=[949.0, 948.0] min_slope=-1.084 mean_slope=-1.066 drop_kdac=-2.167
    row=18 col=18 noisy_transition=[808.0, 807.0] min_slope=-1.015 mean_slope=-1.006 drop_kdac=-1.989
    row=18 col=18 noisy_transition=[671.0, 670.0] min_slope=-1.023 mean_slope=-1.014 drop_kdac=-2.005
    row=18 col=18 noisy_transition=[536.0, 535.0] min_slope=-1.063 mean_slope=-1.055 drop_kdac=-2.091
    row=18 col=18 noisy_transition=[398.0, 397.0] min_slope=-1.047 mean_slope=-1.038 drop_kdac=-2.053
    row=18 col=18 noisy_transition=[257.0, 256.0] min_slope=-1.024 mean_slope=-1.015 drop_kdac=-2.008
    row=18 col=18 noisy_transition=[124.0, 123.0] min_slope=-1.056 mean_slope=-1.047 drop_kdac=-2.074
    row=19 col=18 noisy_transition=[654.0, 652.0] min_slope=-2.549 mean_slope=-1.699 drop_kdac=-5.099
    row=19 col=18 noisy_transition=[350.0, 325.0] min_slope=-2.173 mean_slope=-0.321 drop_kdac=-8.315
    row=19 col=18 noisy_transition=[141.0, 140.0] min_slope=-1.013 mean_slope=-0.939 drop_kdac=-2.020
    row=30 col=18 noisy_transition=[526.0, 524.0] min_slope=-1.259 mean_slope=-0.832 drop_kdac=-2.517
    row=30 col=18 noisy_transition=[392.0, 387.0] min_slope=-2.575 mean_slope=-1.315 drop_kdac=-7.916
    row=30 col=18 noisy_transition=[328.0, 306.0] min_slope=-4.051 mean_slope=-0.521 drop_kdac=-12.007
  Best bias = 400.0
  Max detectors in transition at best bias = 6
  Detectors at best bias in [320.0, 400.0]: [(15, 18), (21, 18), (22, 18), (28, 18), (29, 18), (31, 18)]
    row=15 col=18 transition=[340.0, 320.0] min_slope=-1.372 mean_slope=-0.264 drop_kdac=-5.453
    row=21 col=18 transition=[364.0, 334.0] min_slope=-3.563 mean_slope=-0.383 drop_kdac=-11.886
    row=22 col=18 transition=[385.0, 361.0] min_slope=-2.865 mean_slope=-0.393 drop_kdac=-9.773
    row=28 col=18 transition=[357.0, 329.0] min_slope=-2.681 mean_slope=-0.350 drop_kdac=-10.160
    row=29 col=18 transition=[380.0, 353.0] min_slope=-2.456 mean_slope=-0.326 drop_kdac=-9.126
    row=31 col=18 transition=[386.0, 367.0] min_slope=-3.533 mean_slope=-0.528 drop_kdac=-10.506
  User bias = 500
  Clean detectors in transition in [420.0, 500.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=0 col=18 transition=[536.0, 534.0] min_slope=-0.816 mean_slope=-0.540 drop_kdac=-1.632
gap_to_window=36.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=0 col=18 transition=[408.0, 407.0] min_slope=-0.853 mean_slope=-0.846 drop_kdac=-1.675
gap_to_window=13.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=5 col=18 transition=[540.0, 539.0] min_slope=-0.800 mean_slope=-0.786 drop_kdac=-1.591
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=5 col=18 transition=[514.0, 513.0] min_slope=-0.789 mean_slope=-0.778 drop_kdac=-1.574
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=5 col=18 transition=[488.0, 487.0] min_slope=-0.778 mean_slope=-0.763 drop_kdac=-1.544
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=5 col=18 transition=[463.0, 462.0] min_slope=-0.819 mean_slope=-0.804 drop_kdac=-1.626
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=5 col=18 transition=[437.0, 436.0] min_slope=-0.805 mean_slope=-0.791 drop_kdac=-1.601
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=5 col=18 transition=[411.0, 410.0] min_slope=-0.794 mean_slope=-0.778 drop_kdac=-1.576
gap_to_window=10.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=5 col=18 transition=[386.0, 385.0] min_slope=-0.845 mean_slope=-0.832 drop_kdac=-1.681
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=6 col=18 transition=[522.0, 521.0] min_slope=-0.901 mean_slope=-0.895 drop_kdac=-1.797
```

```
gap_to_window=22.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=6 col=18 transition=[489.0, 488.0] min_slope=-0.750 mean_slope=-0.736 drop_kdac=-1.499
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=6 col=18 transition=[457.0, 456.0] min_slope=-0.734 mean_slope=-0.728 drop_kdac=-1.465
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=6 col=18 transition=[424.0, 423.0] min_slope=-0.724 mean_slope=-0.714 drop_kdac=-1.443
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=6 col=18 transition=[392.0, 391.0] min_slope=-0.735 mean_slope=-0.730 drop_kdac=-1.466
gap_to_window=29.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=7 col=18 transition=[437.0, 421.0] min_slope=-0.619 mean_slope=-0.201 drop_kdac=-3.006
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=8 col=18 transition=[420.0, 419.0] min_slope=-0.087 mean_slope=-0.086 drop_kdac=-0.097
gap_to_window=1.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=8 col=18 transition=[416.0, 403.0] min_slope=-0.124 mean_slope=-0.101 drop_kdac=-1.349
gap_to_window=17.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=8 col=18 transition=[397.0, 396.0] min_slope=-0.525 mean_slope=-0.451 drop_kdac=-0.400
gap_to_window=24.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=18 col=18 transition=[536.0, 535.0] min_slope=-1.063 mean_slope=-1.055 drop_kdac=-2.091
gap_to_window=36.0 reason=excluded as noisy (multiple accepted transitions)
    row=18 col=18 transition=[398.0, 397.0] min_slope=-1.047 mean_slope=-1.038 drop_kdac=-2.053
gap_to_window=23.0 reason=excluded as noisy (multiple accepted transitions)
    row=23 col=18 transition=[414.0, 409.0] min_slope=-0.090 mean_slope=-0.082 drop_kdac=-0.445
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=23 col=18 transition=[403.0, 383.0] min_slope=-0.347 mean_slope=-0.163 drop_kdac=-3.200
gap_to_window=37.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=23 col=18 transition=[380.0, 380.0] min_slope=-0.103 mean_slope=-0.103 drop_kdac=0.000
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=24 col=18 transition=[382.0, 381.0] min_slope=-0.101 mean_slope=-0.096 drop_kdac=-0.118
gap_to_window=39.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=30 col=18 transition=[526.0, 524.0] min_slope=-1.259 mean_slope=-0.832 drop_kdac=-2.517
gap_to_window=26.0 reason=excluded as noisy (multiple accepted transitions)
    row=30 col=18 transition=[392.0, 387.0] min_slope=-2.575 mean_slope=-1.315 drop_kdac=-7.916
gap_to_window=33.0 reason=excluded as noisy (multiple accepted transitions)

Column 19
  Noisy detectors excluded: []
  Best bias = 243.0
  Max detectors in transition at best bias = 6
  Detectors at best bias in [163.0, 243.0]: [(3, 19), (4, 19), (7, 19), (11, 19), (15, 19), (22, 19)]
    row=3 col=19 transition=[191.0, 171.0] min_slope=-2.042 mean_slope=-0.367 drop_kdac=-7.680
    row=4 col=19 transition=[217.0, 200.0] min_slope=-3.064 mean_slope=-0.495 drop_kdac=-8.883
    row=7 col=19 transition=[182.0, 163.0] min_slope=-3.744 mean_slope=-0.554 drop_kdac=-11.057
    row=11 col=19 transition=[235.0, 215.0] min_slope=-2.378 mean_slope=-0.370 drop_kdac=-7.781
    row=15 col=19 transition=[187.0, 168.0] min_slope=-3.991 mean_slope=-0.587 drop_kdac=-11.722
```

```
    row=22 col=19 transition=[191.0, 173.0] min_slope=-2.239 mean_slope=-0.383 drop_kdac=-7.376
  User bias = 700
  Clean detectors in transition in [620.0, 700.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 20
  Noisy detectors excluded: []
  Best bias = 336.0
  Max detectors in transition at best bias = 5
  Detectors at best bias in [256.0, 336.0]: [(1, 20), (5, 20), (8, 20), (12, 20), (15, 20)]
    row=1 col=20 transition=[279.0, 256.0] min_slope=-3.892 mean_slope=-0.473 drop_kdac=-11.491
    row=5 col=20 transition=[287.0, 265.0] min_slope=-1.154 mean_slope=-0.256 drop_kdac=-5.869
    row=8 col=20 transition=[320.0, 296.0] min_slope=-1.277 mean_slope=-0.238 drop_kdac=-5.936
    row=12 col=20 transition=[330.0, 308.0] min_slope=-1.381 mean_slope=-0.271 drop_kdac=-6.246
    row=15 col=20 transition=[284.0, 263.0] min_slope=-1.707 mean_slope=-0.316 drop_kdac=-6.921
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=24 col=20 transition=[40.0, 34.0] min_slope=-0.257 mean_slope=-0.162 drop_kdac=-0.962
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=25 col=20 transition=[2.0, 1.0] min_slope=-1.157 mean_slope=-0.860 drop_kdac=-1.157 gap_to_window=2.0
reason=accepted transition outside user window
    row=30 col=20 transition=[36.0, 35.0] min_slope=-0.660 mean_slope=-0.659 drop_kdac=-1.314
gap_to_window=36.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 21
  Noisy detectors excluded: []
  Best bias = 296.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [216.0, 296.0]: [(11, 21), (12, 21), (20, 21)]
    row=11 col=21 transition=[267.0, 249.0] min_slope=-1.015 mean_slope=-0.204 drop_kdac=-3.927
    row=12 col=21 transition=[240.0, 218.0] min_slope=-2.621 mean_slope=-0.384 drop_kdac=-8.865
    row=20 col=21 transition=[238.0, 216.0] min_slope=-1.383 mean_slope=-0.289 drop_kdac=-6.635
  User bias = 250
  Clean detectors in transition in [170.0, 250.0]: [(12, 21), (20, 21)]
    row=12 col=21 transition=[240.0, 218.0] min_slope=-2.621 mean_slope=-0.384 drop_kdac=-8.865
    row=20 col=21 transition=[238.0, 216.0] min_slope=-1.383 mean_slope=-0.289 drop_kdac=-6.635
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=8 col=21 transition=[206.0, 192.0] min_slope=-0.455 mean_slope=-0.160 drop_kdac=-2.452
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=9 col=21 transition=[239.0, 222.0] min_slope=-0.242 mean_slope=-0.135 drop_kdac=-2.344
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=11 col=21 transition=[267.0, 249.0] min_slope=-1.015 mean_slope=-0.204 drop_kdac=-3.927
gap_to_window=17.0 reason=accepted transition outside user window
```

```
    row=13 col=21 transition=[167.0, 150.0] min_slope=-3.169 mean_slope=-0.528 drop_kdac=-9.510
gap_to_window=20.0 reason=accepted transition outside user window
    row=19 col=21 transition=[277.0, 252.0] min_slope=-0.956 mean_slope=-0.216 drop_kdac=-5.629
gap_to_window=27.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 22
  Noisy detectors excluded: []
  Best bias = 889.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [809.0, 889.0]: [(5, 22), (7, 22), (10, 22)]
    row=5 col=22 transition=[814.0, 809.0] min_slope=-2.813 mean_slope=-0.994 drop_kdac=-6.082
    row=7 col=22 transition=[863.0, 857.0] min_slope=-2.761 mean_slope=-0.860 drop_kdac=-6.179
    row=10 col=22 transition=[883.0, 880.0] min_slope=-1.719 mean_slope=-0.900 drop_kdac=-3.641
  User bias = 250
  Clean detectors in transition in [170.0, 250.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=3 col=22 transition=[211.0, 210.0] min_slope=-0.118 mean_slope=-0.109 drop_kdac=-0.207
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=19 col=22 transition=[215.0, 212.0] min_slope=-0.251 mean_slope=-0.182 drop_kdac=-0.604
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=22 transition=[229.0, 228.0] min_slope=-0.607 mean_slope=-0.599 drop_kdac=-1.191
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=27 col=22 transition=[178.0, 177.0] min_slope=-0.616 mean_slope=-0.580 drop_kdac=-1.110
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 23
  Noisy detectors excluded: []
  Best bias = 324.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [244.0, 324.0]: [(1, 23), (4, 23)]
    row=1 col=23 transition=[258.0, 252.0] min_slope=-1.071 mean_slope=-0.408 drop_kdac=-2.825
    row=4 col=23 transition=[264.0, 244.0] min_slope=-1.145 mean_slope=-0.257 drop_kdac=-5.456
  User bias = 1250
  Clean detectors in transition in [1170.0, 1250.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 24
  Noisy detectors excluded: []
  Best bias = 372.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [292.0, 372.0]: [(18, 24)]
    row=18 col=24 transition=[309.0, 292.0] min_slope=-2.170 mean_slope=-0.391 drop_kdac=-7.112
  User bias = 950
  Clean detectors in transition in [870.0, 950.0]: []
```

```
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 25
  Noisy detectors excluded: []
  Best bias = 480.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [400.0, 480.0]: [(21, 25)]
    row=21 col=25 transition=[406.0, 400.0] min_slope=-1.642 mean_slope=-1.119 drop_kdac=-7.866
  User bias = 1500
  Clean detectors in transition in [1420.0, 1500.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=16 col=25 transition=[1501.0, 1500.0] min_slope=-0.122 mean_slope=-0.104 drop_kdac=-0.170
gap_to_window=1.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=16 col=25 transition=[1487.0, 1487.0] min_slope=-0.112 mean_slope=-0.112 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=16 col=25 transition=[1481.0, 1481.0] min_slope=-0.210 mean_slope=-0.210 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

Column 26
  Noisy detectors excluded: []
  Best bias = 363.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [283.0, 363.0]: [(1, 26)]
    row=1 col=26 transition=[308.0, 283.0] min_slope=-2.015 mean_slope=-0.289 drop_kdac=-7.532
  User bias = 750
  Clean detectors in transition in [670.0, 750.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=12 col=26 transition=[739.0, 739.0] min_slope=-0.134 mean_slope=-0.134 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[736.0, 736.0] min_slope=-0.082 mean_slope=-0.082 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[722.0, 722.0] min_slope=-0.092 mean_slope=-0.092 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[719.0, 719.0] min_slope=-0.081 mean_slope=-0.081 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[715.0, 715.0] min_slope=-0.120 mean_slope=-0.120 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
```

```
    row=12 col=26 transition=[707.0, 707.0] min_slope=-0.237 mean_slope=-0.237 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[702.0, 701.0] min_slope=-0.154 mean_slope=-0.132 drop_kdac=-0.248
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=12 col=26 transition=[691.0, 690.0] min_slope=-0.084 mean_slope=-0.083 drop_kdac=-0.111
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=12 col=26 transition=[683.0, 682.0] min_slope=-0.155 mean_slope=-0.151 drop_kdac=-0.292
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=12 col=26 transition=[671.0, 671.0] min_slope=-0.101 mean_slope=-0.101 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[636.0, 636.0] min_slope=-0.167 mean_slope=-0.167 drop_kdac=0.000
gap_to_window=34.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[787.0, 787.0] min_slope=-0.088 mean_slope=-0.088 drop_kdac=0.000
gap_to_window=37.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[774.0, 774.0] min_slope=-0.106 mean_slope=-0.106 drop_kdac=0.000
gap_to_window=24.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[745.0, 745.0] min_slope=-0.097 mean_slope=-0.097 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[739.0, 738.0] min_slope=-0.130 mean_slope=-0.111 drop_kdac=-0.195
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 27
  Noisy detectors excluded: [(12, 27), (22, 27)]
    row=12 col=27 noisy_transition=[510.0, 499.0] min_slope=-1.773 mean_slope=-0.453 drop_kdac=-3.657
    row=12 col=27 noisy_transition=[496.0, 493.0] min_slope=-1.310 mean_slope=-0.595 drop_kdac=-1.219
    row=22 col=27 noisy_transition=[542.0, 506.0] min_slope=-2.276 mean_slope=-0.321 drop_kdac=-11.779
    row=22 col=27 noisy_transition=[503.0, 503.0] min_slope=-1.251 mean_slope=-1.251 drop_kdac=0.000
  Best bias = 547.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [467.0, 547.0]: [(28, 27)]
    row=28 col=27 transition=[472.0, 467.0] min_slope=-2.187 mean_slope=-0.951 drop_kdac=-5.673
  User bias = 690
  Clean detectors in transition in [610.0, 690.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=23 col=27 transition=[630.0, 628.0] min_slope=-0.084 mean_slope=-0.081 drop_kdac=-0.151
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=23 col=27 transition=[625.0, 590.0] min_slope=-0.243 mean_slope=-0.128 drop_kdac=-4.432
gap_to_window=20.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=27 transition=[588.0, 588.0] min_slope=-0.097 mean_slope=-0.097 drop_kdac=0.000
```

```
gap_to_window=22.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=27 transition=[691.0, 690.0] min_slope=-0.097 mean_slope=-0.095 drop_kdac=-0.158
gap_to_window=1.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 28
  Noisy detectors excluded: []
  Best bias = 738.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [658.0, 738.0]: [(26, 28)]
    row=26 col=28 transition=[677.0, 658.0] min_slope=-2.012 mean_slope=-0.291 drop_kdac=-5.725
  User bias = 650
  Clean detectors in transition in [570.0, 650.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=26 col=28 transition=[677.0, 658.0] min_slope=-2.012 mean_slope=-0.291 drop_kdac=-5.725
gap_to_window=27.0 reason=accepted transition outside user window

Column 29
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 750
  Clean detectors in transition in [670.0, 750.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=3 col=29 transition=[782.0, 781.0] min_slope=-0.184 mean_slope=-0.153 drop_kdac=-0.311
gap_to_window=32.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=29 transition=[761.0, 760.0] min_slope=-0.237 mean_slope=-0.204 drop_kdac=-0.440
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=29 transition=[739.0, 738.0] min_slope=-0.240 mean_slope=-0.203 drop_kdac=-0.427
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=29 transition=[717.0, 716.0] min_slope=-0.184 mean_slope=-0.142 drop_kdac=-0.278
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=29 transition=[697.0, 696.0] min_slope=-0.219 mean_slope=-0.188 drop_kdac=-0.387
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=29 transition=[677.0, 676.0] min_slope=-0.206 mean_slope=-0.170 drop_kdac=-0.346
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=29 transition=[658.0, 657.0] min_slope=-0.194 mean_slope=-0.151 drop_kdac=-0.287
gap_to_window=13.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=29 transition=[641.0, 640.0] min_slope=-0.247 mean_slope=-0.205 drop_kdac=-0.404
gap_to_window=30.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=29 transition=[785.0, 784.0] min_slope=-0.128 mean_slope=-0.115 drop_kdac=-0.216
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=29 transition=[699.0, 698.0] min_slope=-0.098 mean_slope=-0.096 drop_kdac=-0.168
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
```

```
    row=17 col=29 transition=[670.0, 670.0] min_slope=-0.137 mean_slope=-0.137 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=17 col=29 transition=[642.0, 641.0] min_slope=-0.157 mean_slope=-0.146 drop_kdac=-0.301
gap_to_window=29.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[781.0, 780.0] min_slope=-0.178 mean_slope=-0.150 drop_kdac=-0.322
gap_to_window=31.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[752.0, 751.0] min_slope=-0.158 mean_slope=-0.127 drop_kdac=-0.273
gap_to_window=2.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[724.0, 723.0] min_slope=-0.198 mean_slope=-0.173 drop_kdac=-0.371
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[695.0, 694.0] min_slope=-0.184 mean_slope=-0.156 drop_kdac=-0.337
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[666.0, 665.0] min_slope=-0.167 mean_slope=-0.138 drop_kdac=-0.297
gap_to_window=5.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[638.0, 637.0] min_slope=-0.219 mean_slope=-0.187 drop_kdac=-0.417
gap_to_window=33.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=29 transition=[724.0, 724.0] min_slope=-0.095 mean_slope=-0.095 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=29 transition=[638.0, 638.0] min_slope=-0.097 mean_slope=-0.097 drop_kdac=0.000
gap_to_window=32.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

Column 30
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 31
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Saved PDF:
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/home/timestudent/anie/load_curves_all/specB/best_bias_20260208_spec_B_iv_60_el_349mK_iv_all_data.pdf